

OpenAI Math Digest

CATALOGUE 033

Campana's orbifold Iitaka conjecture and logarithmic subadditivity

Catalogue synthesis & 5 proof overviews

Independent reading guide · English edition

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AI-generated explanations may contain errors. Check important claims and proofs against the original manuscripts. This booklet collects the website's explanations, not the original papers. Expert review and formal verification have not been performed.

Contents

The synthesis comes first, followed by all 5 overviews in official catalogue order. The contents and citations are clickable. Wide tables are arranged as entries for print; the claims and source references are retained.

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Original-manuscript snapshot: `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`

Campana's orbifold Iitaka conjecture and logarithmic subadditivity

The five papers do not form a single chain of Hodge-theoretic inputs. OI supplies a lower bound and adjoint positivity at different stages of WV. RA proves its upper bound independently and adds OI only for additivity. PH gives a separate proof of ordinary subadditivity, while BS concerns semiampleness of the actual moduli line. This synthesis matches the manuscripts' claims with the articles and specifies what each input supplies at its point of use.

5 manuscript summaries **13** selected connections **5** proof overviews

This reading map records 12 inputs and one premise match. Alternatives, similar methods, and background are distinguished from proof dependencies. The arrows are neither a complete dependency graph nor a certification of the proofs.

Articles in official order

The official field is **Algebraic and complex geometry**. The page title follows overview entry 033; CONTENTS uses the alternative heading Iitaka subadditivity, variation, and logarithmic additivity. There are five papers and 259 pages including references. Each title links to its English article.

Order / article

01 Orbifold and logarithmic Iitaka subadditivity OI

Version / pages

2026-09-26 · 55 pages · [Source](#)

Role

Supplies orbifold and logarithmic lower bounds and adjoint positivity of Hodge lines.

Order / article

02 Logarithmic Kodaira dimension and whole-fiber variation WV

Version / pages

2026-09-26 · 75 pages · [Source](#)

Role

Descends the whole fiber to the parameter field, giving a lower bound involving variation.

Order / article

03 The reverse logarithmic Kodaira inequality and additivity RA

Version / pages

2026-09-26 · 51 pages · [Source](#)

Role

Proves the upper bound under stratum smoothness, then combines it with OI's lower bound.

Order / article

04 Projective Hodge lines and ordinary Iitaka subadditivity PH

Version / pages

2026-09-27 · 40 pages · [Source](#)

Role

A separate route from projective Hodge lines to ordinary subadditivity.

Order / article

05 B-semiampleness for compact log-smooth Kähler fibrations BS

Version / pages

2026-09-10 · 38 pages · [Source](#)

Role

Identifies the lct moduli line and proves semiampleness by decomposition, extension and norm descent.

Each paper has Japanese and English articles with proof diagrams. OI and RA retain their main-proof overview diagrams. WV now has a map from the parameter field to whole-fiber descent, and BS one from stabilization to semiampleness. Diagram counts follow the arguments.

[Official overview](#) · [PDF p. 5](#) · [Official CONTENTS](#)

Three strands of the argument

These groups organize the roles of the arguments. The inventory above retains the official catalogue order. Follow each strand to its diagrams and the explanations immediately after them.

01

Lower bounds, variation, upper bounds ↓

Which OI results enter WV and RA

Separate logarithmic subadditivity from adjoint positivity of the Hodge line, then locate the step combining the lower bound with RA's independent upper bound.

- 01 OI
- 02 WV
- 03 RA

02

Relative Iitaka constructions and Kähler induction ↓

Why compare sections, Hodge lines, and boundaries

Track auxiliary inputs to WV's additional route and KA in catalogue 034, distinguishing them from inputs to the main theorem.

- 01 OI
- 02 WV
- 034 KA

03

Good models and alternative routes ↓

Connections with 034 and the roles of PH and BS

Follow OI into LA and CK, and LA into a consequence in WV. PH’s alternative proof and BS’s semiamplessness argument are treated separately in the comparison section.

- 034 LA
- 034 CK
- 04 PH
- 05 BS

The connection between the strands

OI supplies different results to different stages in WV, LA, and KA. LA returns a good model to a consequence in WV. PH and BS have their own arguments; shared Hodge-theoretic language does not by itself establish a direct dependency.

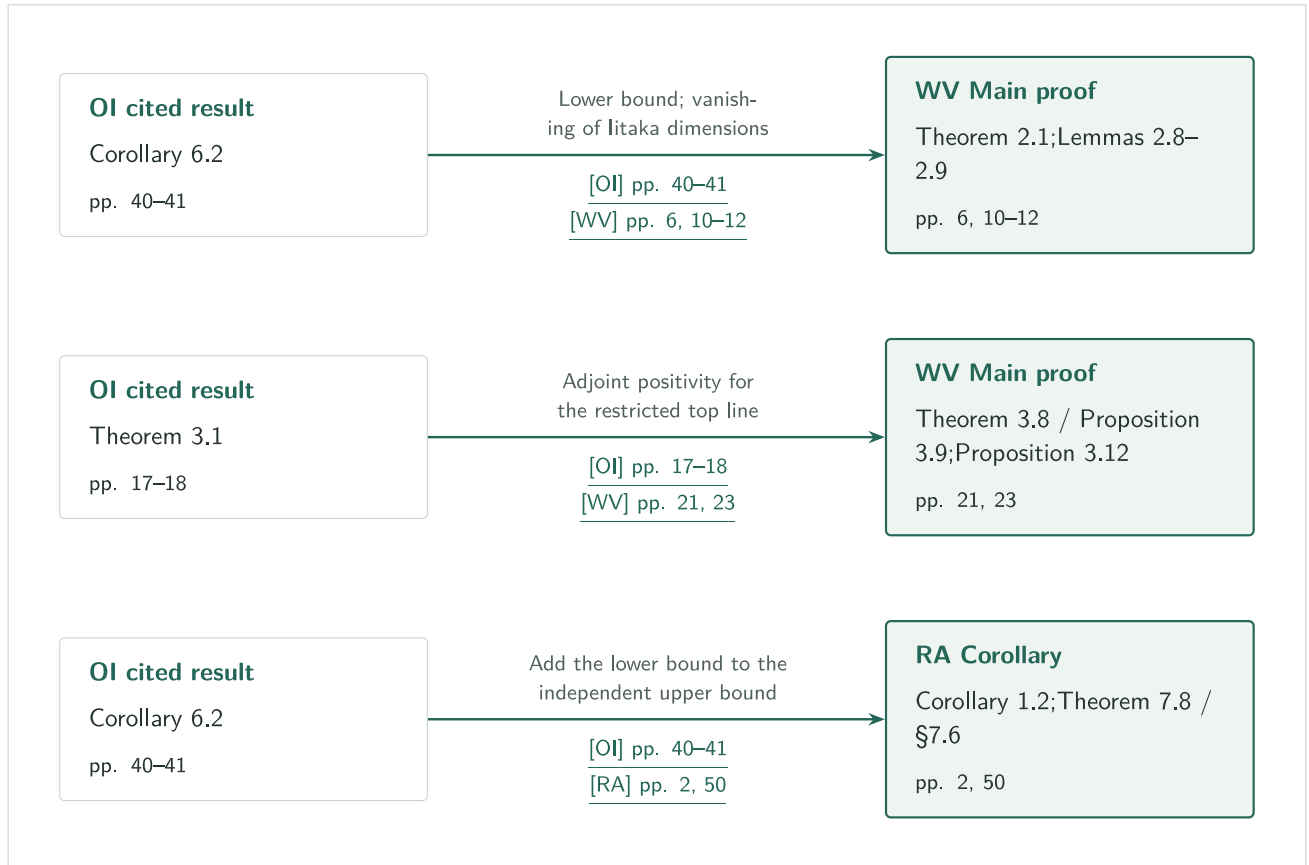
References on the arrows

OI, WV, RA, PH and BS denote the five papers above. In catalogue 034, LA is Log abundance in characteristic zero, KA is Log abundance for compact Kähler spaces under logarithmic Itaka subadditivity, and CK is Conditional good minimal models for compact Kähler fourfolds.

Solid arrows record inputs actually used by the specified receiving result. Corollary and additional-result inputs are not moved into the main proof. Dashed arrows match an explicitly stated premise. Alternative proofs, similar methods and background are discussed separately, without dependency arrows. Citations link to pinned sources; page numbers are given in the labels.

From lower bounds to variation, and then to additivity

Goal. Separate WV’s two OI inputs from the OI input used after RA’s upper bound is complete. The two WV boxes represent different stages of one main proof; the RA box represents the additivity corollary.



The two direct OI inputs into WV and the lower-bound input into RA additivity

[Open figure ↗](#) [TeX source ↓](#)

1. Measure the parameter field using logarithmic subadditivity

WV restates OI Corollary 6.2 as Theorem 2.1. Compatible projective SNC compactifications satisfy the inverse-boundary support condition and give $\bar{\kappa}(U) \geq d + \bar{\kappa}(V)$ when $d = \kappa(F) \geq 0$. On an Iitaka fiber, two nonnegative Kodaira dimensions must then vanish. Proportional logarithmic forms measure the family of images. Proposition 2.10 produces $\mathbb{C}(T_0) \subseteq \mathbb{C}(Y)$ and $\bar{\kappa}(U) = d + \dim T_0$. Constancy of the whole fiber has not yet been used.

[OI, Corollary 6.2 · pp. 40–41](#) · [WV, Theorem 2.1, Lemmas 2.8–2.9, Proposition 2.10 · pp. 6, 10–14](#)

Connection details: [OI](#) → [WV](#) · [Corollary 6.2](#)

[OI](#) · [Source result in the overview](#) → [WV](#) · [Application in the overview](#) →

2. Use the restricted Hodge line for the first constancy step

OI Theorem 3.1 supplies adjoint positivity for a complex summand of an integral ambient variation. WV has an entire highest line and reapplies the theorem to the variation restricted to an Iitaka fiber. The upper bound on the section dimension of $K + B + M$, together with nonvanishing of $K + B$, excludes a positive-dimensional adjoint base and makes M rationally

trivial. Finite character, analytic birational constancy, markings, cocycles and descent of finite covers remain distinct subsequent steps. Period-image dimension is not identified with whole-fiber variation.

OI, Theorem 3.1 · pp. 17–18 ·

WV, Theorem 3.8, Propositions 3.9, 3.12, Corollary 3.16 · pp. 21–25; §§4–7 · pp. 25–59

Connection details:OI → WV · Theorem 3.1

OI · Source result in the overview → WV · Application in the overview →

3. Add the lower bound for the same pair to RA’s upper bound

RA Theorem 1.1 proves an upper bound when all boundary strata are smooth over the open base. OI enters only in Corollary 1.2: substitute $D_X = E, D_Y = D$ into Corollary 6.2 for the same pair and very general fiber. Finite bounds give equality. If a term on the right is $-\infty$, vanishing in every positive degree comes from RA’s upper bound.

RA, Theorem 1.1, Corollary 1.2 · p. 2; Theorem 7.8, §7.6 · p. 50 · OI, Corollary 6.2 · pp. 40–41

Use of the output. Whole-function-field descent proves $\text{Var}(f) \leq \dim T_0$ for WV. Combining this with the dimension identity gives its main inequality. A separate route compares RA additivity with WV Theorem 1.1: for a smooth family with $\kappa(F) \geq 0$ and $\bar{\kappa}(V) \geq 0$, it yields $\text{Var}(f) \leq \bar{\kappa}(V)$. This does not cover the negative-base branch or every special base.

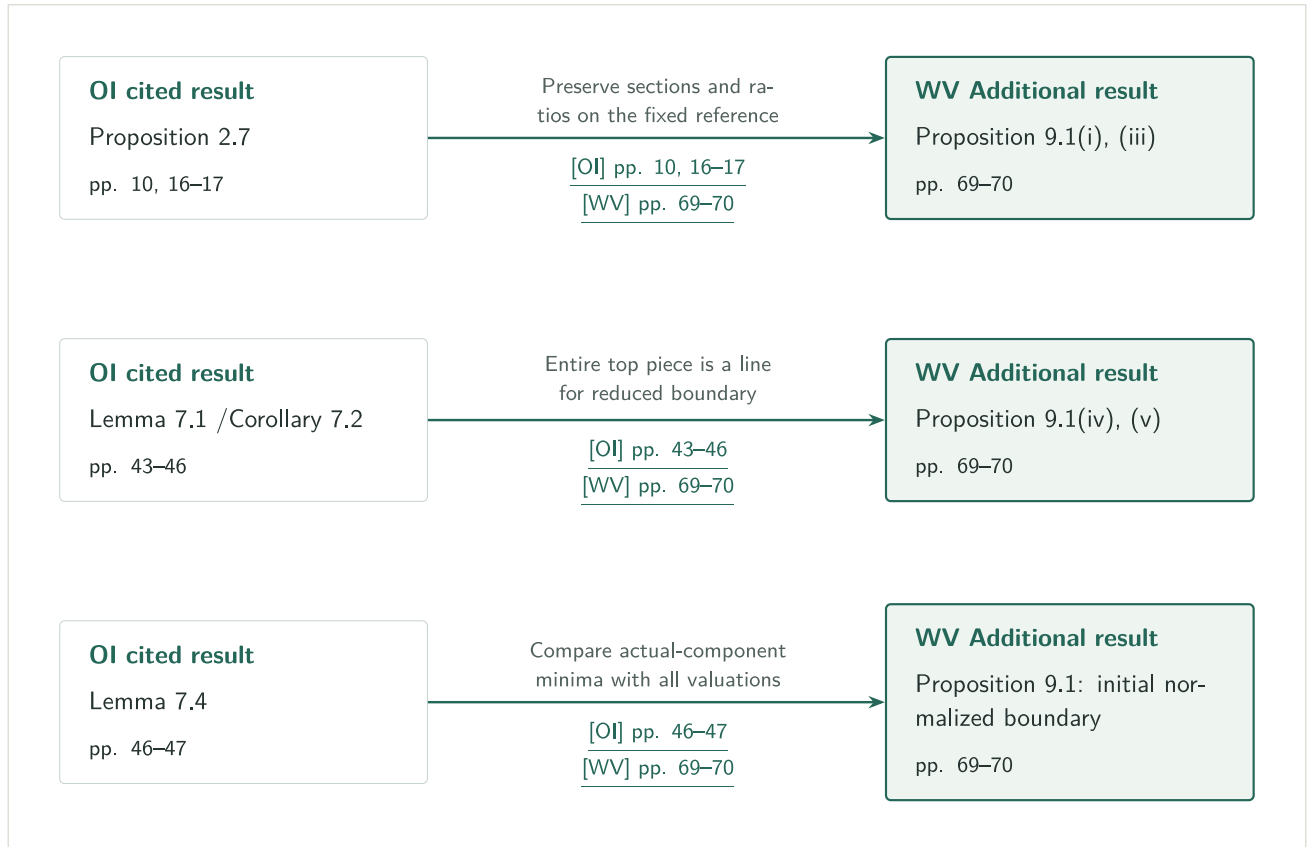
WV, §7.4 · pp. 58–59; comparison after Corollary 1.3 · p. 5

Connection details:OI → RA · Corollary 6.2

OI · Source result in the overview → RA · Application in the overview →

Three inputs for the additional relative Iitaka construction

Goal. WV §§8–9 form an additional route. Retain the section ring of the original X while obtaining the entire highest line of a pure integral variation on the ordinary relative Iitaka base. The diagram separates three inputs to Proposition 9.1.



OI inputs for section comparison, the entire highest line and boundary normalization in WV Proposition 9.1

[Open figure ↗](#) [TeX source ↓](#)

1. Return sections and ratios to the fixed reference space

WV uses OI Proposition 2.7 with the original X as reference. Preservation of section-image dimension on the generic fiber gives both bigness of $K_W + B + M$ over Y and comparison with the original pluricanonical systems.

OI, Proposition 2.7 · pp. 10, 16–17 · WV, Proposition 9.1 (i), (iii) and proof · pp. 69–70

Connection details: OI → WV · Proposition 2.7

OI · Source result in the overview → WV · Application in the overview →

2. Obtain the entire highest line for a reduced boundary

OI Lemma 7.1 and Corollary 7.2 use the least-index root cover of a reduced SNC pair of logarithmic Kodaira dimension zero. The entire highest piece, not only a character eigenspace, is rank one; M is identified with its parabolic extension. Reducedness is essential and cannot be dropped for general rational coefficients.

OI, Lemma 7.1, Corollary 7.2, Remark 7.3 · pp. 43–46 · WV, Proposition 9.1 (iv), (v) · pp. 69–70

Connection details: OI → WV · Lemma 7.1 / Corollary 7.2

[OI · Source result in the overview](#) → [WV · Application in the overview](#) →

3. Normalize the boundary while distinguishing the two minima

OI Lemma 7.4 compares the minimum over actual source components with the lct allowing valuations on higher models, on the same SNC preparation. WV uses it to normalize the initial boundary. This synthesis compared the statement and receiving use; the full valuation and model-change arguments were not independently verified.

[OI, Lemma 7.4 · pp. 46–47](#) · [WV, proof of Proposition 9.1 · p. 70](#)

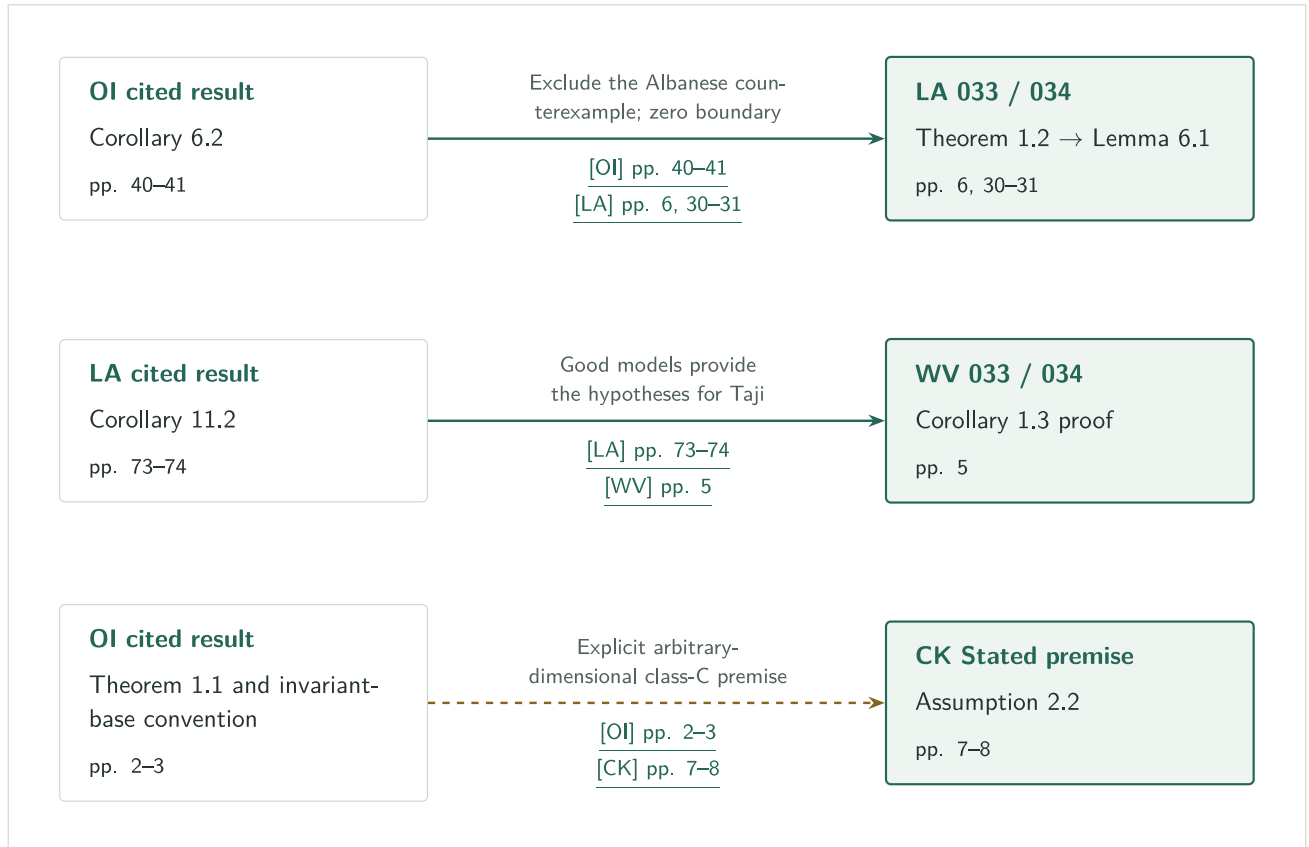
Use of the output. The data enter WV Theorem 8.1. Corollaries 9.2–9.4 connect the second Itaka base to the definition field of the original whole fiber. Intermediate volume and degree estimates in §8 and the complete argument of §9 remain only partially studied.

Connection details: [OI](#) → [WV · Lemma 7.4](#)

[OI · Source result in the overview](#) → [WV · Application in the overview](#) →

Connect models in 034 with consequences in 033

Goal. The uses of OI in 034 differ from the return input from LA into WV. The dashed CK edge records a match to an explicit premise.



OI inputs into LA and the CK premise, and LA input into WV Corollary 1.3

[Open figure ↗](#) [TeX source ↓](#)

1. LA uses subadditivity only in its Albanese reduction

Under the lower-dimensional good-model induction hypothesis, LA Lemma 6.1 considers the Albanese map of a hypothetical nonvanishing counterexample. The smooth base maps generically finitely to a subvariety of an abelian variety and has nonnegative Kodaira dimension. The fiber has nonnegative Kodaira dimension by induction. OI Corollary 6.2 with zero boundaries contradicts nonvanishing failure. This is LA’s sole use of subadditivity. LA p.6 explicitly records an alternative using Hacon–Popa–Schnell for this particular application.

LA, Theorem 1.2 · p. 6; Lemma 6.1 · pp. 30–31 · OI, Corollary 6.2 · pp. 40–41

Connection details: OI → LA · Corollary 6.2

OI · [Source result in the overview](#) → LA · [Application in the overview](#) →

2. WV’s smooth-family corollary takes good models as input

In WV Corollary 1.3, BDPP turns non-uniruledness of each closed fiber into pseudo-effectivity of K_F . LA Corollary 11.2 supplies a semiample model and an effective exceptional comparison. The comparison gives canonical singularities, permitting Taji’s theorem to yield the two variation branches and the special-base conclusion. LA is not an input to WV Theorem 1.1.

LA, Corollary 11.2 · pp. 73–74 · WV, Corollary 1.3 and proof · p. 5

Connection details: [LA](#) → [WV](#) · [Corollary 11.2](#)

[LA](#) · [Source result in the overview](#) → [WV](#) · [Application in the overview](#) →

3. Preserve CK's conditional framework

CK Assumption 2.2 explicitly requires arbitrary-dimensional class-C orbifold subadditivity, including coefficient one and the invariant-base/neat-model conventions. OI Theorem 1.1 and its base definition match that premise. CK's other assumptions on the MMP and abundance after nonvanishing are not thereby discharged.

[OI, definitions and Theorem 1.1](#) · pp. 2–3 · [CK, Assumption 2.2, \(2\)–\(5\)](#) · pp. 7–8

Use of the output. LA's good model is also used by Schnell, whose direct input is LA Corollary 11.2. The path $\text{OI} \rightarrow \text{LA} \rightarrow \text{Schnell}$ is not collapsed to a direct $\text{OI} \rightarrow \text{Schnell}$ edge. The versions and conditional scope of the existing 034 articles are retained.

Connection details: [OI](#) → [CK](#) · [Theorem 1.1 and invariant-base convention](#)

[OI](#) · [Source result in the overview](#) → [CK](#) · [Application in the overview](#) →

Supply specific auxiliary results to the Kähler paper

Goal. The pinned KA manuscript uses separate OI results for reduction, comparison and positivity. The aim is to identify these uses rather than label all of them consequences of subadditivity.



Direct OI inputs into KA's relative litaka reduction, stable-family comparison, Hodge line and effectivity

[Open figure ↗](#) [TeX source ↓](#)

1. Reduce to a Kodaira-zero general fiber

KA §5.1 uses OI Lemma 2.6 to pass to the relative litaka base and return to the lower-dimensional Proposition 5.1. The strict-plus-reduced-exceptional boundary convention preserves the logarithmic Kodaira-zero condition needed for subsequent comparisons.

[OI, Lemma 2.6 · p. 9](#) · [KA, §5.1 · p. 91](#); [Lemma 5.7 · p. 98](#)

Connection details: [OI → KA · Lemma 2.6](#)

[OI · Source result in the overview →](#) [KA · Application in the overview →](#)

2. Exclude the big case by stable-family comparison

Slightly lowering horizontal coefficients preserves fiberwise bigness. KA applies OI Lemma 5.2 to compare with a projective stable family. Algebraic dimension zero forces its projective parameter base to be a point; rational functions on the positive-dimensional projective factor then give a contradiction.

[OI, Lemma 5.2 · p. 32](#) · [KA, §5.1 · p. 91](#)

Connection details: [OI](#) → [KA](#) · [Lemma 5.2](#)

[OI](#) · [Source result in the overview](#) → [KA](#) · [Application in the overview](#) →

3. Compare actual lines and section spaces together

KA Lemma 5.2 combines OI Proposition 2.7 with Theorem 3.1. The output is an identity of rational lines $M = p^*P_S$ and bigness of $K_S + a_0P_S$, not only an equality of numerical classes. Descent to the original reference space gives the comparison of complete section spaces. No additional sign assertion is made here for residual terms mapping into codimension at least two.

[OI, Proposition 2.7, Theorem 3.1 · pp. 10–18](#) · [KA, Lemma 5.2 · pp. 91–94](#)

Connection details: [OI](#) → [KA](#) · [Proposition 2.7 / Theorem 3.1](#)

[OI](#) · [Source result in the overview](#) → [KA](#) · [Application in the overview](#) →

4. Produce a section for interpolation using a big base twist

At p.99, KA Lemma 5.7 explicitly checks OI Lemma 3.3’s nonzero relative system, SNC boundary and projective-base hypotheses. Rational interpolation combines the resulting effective divisor with a negative ample twist and another with a positive twist, forcing positive Kodaira dimension.

[OI, Lemma 3.3 · pp. 18–19](#) · [KA, Lemma 5.7 · pp. 98–99](#)

Use of the output. These results enter KA’s dimension induction and adjoint calculations on the base. KA Assumption 1.1 and the other external inputs remain part of the argument; OI alone is not asserted to imply the whole paper.

Connection details: [OI](#) → [KA](#) · [Lemma 3.3](#)

[OI](#) · [Source result in the overview](#) → [KA](#) · [Application in the overview](#) →

Follow a result into its application

These twelve interfaces cover main proofs, corollaries and additional results and use the same records as the diagrams. Input statements, receiving uses and corresponding articles were compared; full input proofs were not independently verified. The CK premise, three alternatives, two method comparisons and one background relation are kept outside this solid-arrow list.

13 selected connections

INPUT · 033 / 01

OI

Corollary 6.2 · pp. 40–41 ↗

Direct input

Supplies the logarithmic lower bound and forces two nonnegative terms on an Itaka fiber to vanish, enabling the parameter-field construction.

USED IN · 033 / 02

WV

Theorem 2.1; Lemmas 2.8–2.9 · pp. 6, 10–12 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Theorem 3.1 · pp. 17–18 ↗

Direct input

Gives adjoint positivity for the restricted Hodge line. The section bound forces its adjoint base to be a point and makes M rationally trivial.

USED IN · 033 / 02

WV

Theorem 3.8 / Proposition 3.9; Proposition 3.12 · pp. 21, 23 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Corollary 6.2 · pp. 40–41 ↗

Input to a consequence

Supplies the lower bound for the same pair and very general fiber; combining it with RA 's upper bound gives additivity.

USED IN · 033 / 03

RA

Corollary 1.2; Theorem 7.8 / §7.6 · pp. 2, 50 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Proposition 2.7 · pp. 10, 16–17 ↗

Input to an additional result

Preserves relative and absolute section systems and their ratios on the fixed original X , giving relative bigness in the additional ordinary construction.

USED IN · 033 / 02

WV

Proposition 9.1(i), (iii) · pp. 69–70 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Lemma 7.1 / Corollary 7.2 · pp. 43–46 ↗

Input to an additional result

For a reduced boundary, the least-index root cover makes the entire top piece rank one and realizes the normalized M as a line in a pure integral variation.

USED IN · 033 / 02

WV

Proposition 9.1(iv), (v) · pp. 69–70 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Lemma 7.4 · pp. 46–47 ↗

Input to an additional result

Normalizes B by comparing the minimum over actual source components with the lct over all valuations, on the same model as the section and Hodge-line comparison.

USED IN · 033 / 02

WV

Proposition 9.1: initial normalized boundary · pp. 69–70 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Corollary 6.2 · pp. 40–41 ↗

Direct input

Applied with zero boundaries to the Albanese fibration, it forces irregularity zero for a nonvanishing counterexample; this is LA's sole use of subadditivity.

USED IN · 034 / 04

LA

Theorem 1.2 → Lemma 6.1 · pp. 6, 30–31 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Lemma 2.6 · pp. 9 ↗

Direct input

Passes to the relative Iitaka base with Kodaira-zero general fiber so that lower-dimensional induction can continue.

USED IN · 034 / 01

KA

§5.1; reused in Lemma 5.7 · pp. 91, 98 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Lemma 5.2 · pp. 32 ↗

Direct input

Compares the big fibers, after lowering the horizontal boundary, with a projective stable family, excluding the big case at algebraic dimension zero.

USED IN · 034 / 01

KA

§5.1 · pp. 91 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Proposition 2.7 / Theorem 3.1 · pp. 10–17, 17–18 ↗

Direct input

Compares section spaces and the actual rational Hodge line, giving $M=p^*P$ and adjoint bigness on the projective base; signs above codimension two require separate work.

USED IN · 034 / 01

KA

Lemma 5.2 · pp. 91–94 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Lemma 3.3 · pp. 18–19 ↗

Direct input

Produces a section after a big base twist; interpolation with another effective divisor forces positive Kodaira dimension.

USED IN · 034 / 01

KA

Lemma 5.7, interpolation · pp. 98–99 ↗

[Source overview](#) → [Receiving overview](#) →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 034 / 04

LA

Corollary 11.2 · pp. 73–74 ↗

Input to a consequence

Provides a semiample good model and effective exceptional comparison for each closed fiber whose canonical class is pseudo-effective by BDPP; canonicity then permits Taji's theorem.

USED IN · 033 / 02

WV

Corollary 1.3 proof · pp. 5 ↗

Source overview → Receiving overview →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

INPUT · 033 / 01

OI

Theorem 1.1 and invariant-base convention · pp. 2–3 ↗

Premise match

Matches full class-C orbifold subadditivity, including coefficient one and arbitrary dimensions, to the recipient's explicit premise.

USED IN · 034 / 03

CK

Assumption 2.2 · pp. 7–8 ↗

Source overview → Receiving overview →

Statement and receiving use compared; complete source and receiving proofs have not been independently verified.

[Link to this connection](#)

PH and BS, alternatives, and method comparisons

1. PH gives a separate route to ordinary subadditivity

PH uses BBT's full period image and BC's logarithmic general type in Theorem 1.2. It combines Fujino–Mori/Ambro canonical-bundle comparisons, Fujino weak positivity and Hashizume's general-type-fiber subadditivity in §§6–7. Result numbers, versions and uses appear in [PH article §5](#). Theorem 1.1 and OI Corollary 6.3 are separate proofs of ordinary subadditivity.

PH Lemma 8.4/Theorem 8.1 is an alternative adjoint comparison in WV Remark 3.17. Tensor replacement must be performed anew on the restricted variation; no common exponent is required. This is distinct from WV's chosen direct comparison.

[PH, Theorems 1.1–1.2 · p. 2; Theorem 8.1, Lemma 8.4 · pp. 28–29](#) · [WV, Remark 3.17 · p. 25](#)

2. BS uses different inputs to obtain semiampleness

BS uses Fujino–Fujisawa extension for the moduli-line identification in §§3–4, and Matsumura–Wang–Wu–Zhang decomposition plus Toma compactness for the family in §5. In §§6–7, BFMT's algebraic b-semiampleness is applied to the projective comparison factor, while arithmetic period compactifications handle torus and symplectic factors. Norms extend the specified isomorphism, and descent gives global generation at every point. Exact inputs and hypotheses are in [BS article §7](#).

BS p.3 explicitly says that the orbifold Iitaka theorem is not used. No direct edge to another 033 paper has been added here; this is not an exhaustive assertion of independence.

[BS, Theorem 1.1 and proof outline · p. 3; §§5–9 · pp. 18–37](#)

3. Separate similar methods from background

OI and PH both use weak positivity and rational interpolation, but OI's complex summand and PH's entire highest line have different hypotheses. OI and BS both track root eigenlines, boundary orders and actual line bundles, but the orbifold inf-multiplicity base differs from the lct discriminant. These are editorial comparisons of methods, not dependency claims.

RA's article treats the Ax–Schanuel source for the connection-form argument as methodological background: supplying the needed semisimple argument in the manuscript is distinguished from taking an external theorem directly as input.

[OI, Proposition 3.4 · pp. 19–20](#) · [PH, §7.3 · pp. 25–26](#) · [BS, §§3–4 · pp. 6–18](#) · [RA, §5.1, Lemma 5.1 · pp. 26–28](#)

Source versions and remaining checks

The 033 commit pinned on 2026-10-08 is `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. The 034 recipients retain the published articles' commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`: LA 2026-09-24, KA 2026-10-04 and CK 2026-10-05. The October 6 KA/CK revisions are outside this synthesis. Cached PDFs for all five 033 papers and these three 034 papers were rehashed against the inventories.

Article

OI

Principal remaining checks

Complete rank-loss and stable-family arguments, full BFMT interfaces and valuation comparisons in §7, and core corollaries. Checking Lemma 7.4's statement and use does not verify its full proof.

Article

WV

Principal remaining checks

Full regularization and L^2 estimates, original Hanamura and SGA1 purity statements, intermediate §8 estimates and complete §9. The additional OI statements and uses were compared in this synthesis.

Article

RA

Principal remaining checks

Homogeneous replacement, compact flags, finite monodromy, all boundary estimates and representation theory, and full external proofs.

Article

PH

Principal remaining checks

Multivariable Hodge extensions, all auxiliary proofs, and the alternative arguments in §§8–9. The WV alternative is a statement/use comparison.

Article

BS

Principal remaining checks

Complete parameter-space constructions, polarization changes and extension compatibility, and external arithmetic-quotient/compactification theory.

The articles' external-input tables distinguish statement/application comparisons from recipient-only readings. This synthesis retains those limits. Unlisted relations are unexamined or unrecorded; the diagrams do not claim to exhaust all dependencies.

[OpenAI Research Catalog · 033 · p. 5 ↗](#)

[Official catalogue at the reference commit ↗](#)

CONTENTS.md: [litaka subadditivity, variation, and logarithmic additivity](#)

Source snapshot: [fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb](#)

[Catalogue 034: abundance and effective litaka fibrations →](#)

Orbifold and logarithmic Iitaka subadditivity

Exact section comparison on the relative Iitaka base and two positivity inputs

The manuscript asserts orbifold subadditivity for rational SNC pairs in Fujiki class C, including boundary coefficient one. Its argument retains the original logarithmic pluricanonical sections on a relative Iitaka base and combines adjoint positivity of a Hodge line with addition for log-general-type fibers. A fixed section removes the final ample twist by multiplication.

Manuscript author: OpenAI · 2026-09-26 · 55 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

Theorem 1.1 — Orbifold subadditivity

Let X be a smooth compact connected complex manifold in Fujiki class C, and let Δ be an SNC boundary with coefficients in $\mathbb{Q} \cap [0, 1]$. Let $f : X \rightarrow Y$ be a surjective holomorphic map with connected fibers onto a normal compact irreducible complex space. For a very general smooth fiber F and $\Delta_F = \Delta|_F$, the manuscript proves

$$\kappa(X, K_X + \Delta) \geq \kappa(F, K_F + \Delta_F) + \kappa(f| \Delta).$$

A point has Iitaka dimension zero, and $(-\infty) + a = -\infty$, including $a = -\infty$.

On a smooth-base model, put

$$m_\Delta(E) = \frac{1}{1 - \Delta_E}, \quad m(f, \Delta; D) = \min_{E \rightarrow D} \text{ord}_E(f^*D) m_\Delta(E),$$

$$B(f, \Delta) = \sum_D \left(1 - \frac{1}{m(f, \Delta; D)} \right) D.$$

Here $1/0 = \infty$ and $1/\infty = 0$; no divisibility condition is imposed. The base term in the theorem is the invariant

$$\kappa(f \mid \Delta) = \inf_{f' \sim f} \kappa(Y', K_{Y'} + B(f', \Delta'))$$

over the orbifold birational equivalence allowed by (1.2), rather than the dimension of the orbifold base on an arbitrary model. For a normal singular base, resolve the base and the main component, using the strict transform of Δ plus the reduced divisor exceptional over the original source.

Theorem 1.1 · (1.1)–(1.4) · pp. 2–3

Corollary 6.2 — Logarithmic subadditivity

Let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective complex varieties. Let D_X, D_Y be reduced SNC divisors, allowing zero, and assume

$$\text{Supp}(f^*D_Y) \subseteq \text{Supp}(D_X).$$

For a very general fiber F and $D_F = D_X|_F$,

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y).$$

The same statement holds for holomorphic fibrations between smooth compact manifolds in Fujiki class C.

Corollary 6.2 · pp. 40–41

Corollary 6.3 — Ordinary subadditivity in characteristic zero

Let k be an algebraically closed field of characteristic zero, and let $f : X \rightarrow Y$ be a surjective morphism with connected fibers between smooth connected projective k -varieties. For its geometric generic fiber F ,

$$\kappa(X) \geq \kappa(F) + \kappa(Y).$$

More generally, the logarithmic inequality holds over k under the same boundary hypotheses as Corollary 6.2, again replacing a very general complex fiber by the geometric generic fiber.

Corollary 6.3 · p. 41

References on the arrows

Unprefixed result numbers refer to this manuscript [OI](#). The main external inputs use the following keys.

- [Fuj](#) Fujino: weak positivity. Author version 0.54 (June 30, 2015).
- [FF](#) Fujino–Fujisawa: canonical extensions. Author version 0.55 (March 11, 2025).
- [Vil](#) Villadsen: compactification. arXiv v1.
- [BC](#) Brunebarbe–Cadorel: logarithmic general type. arXiv v1.
- [BT](#) Bakker–Tsimmerman: Ax–Schanuel. arXiv v1.
- [KP](#) Kovács–Patakfalvi: stable-family positivity. The 62-page author manuscript.
- [Cam](#) Campana: the orbifold base. arXiv v8.

Page locators are displayed in the links; GitHub PDF previews are not assumed to navigate to a specified page automatically.

2. Proof of Theorem 1.1 — Transfer to the relative Iitaka base

Proof route. The main proof combines exact section comparison on the relative Iitaka base with two positivity inputs. The next section expands the final [cancellation of the ample twist](#); the resulting lower bound then yields [logarithmic subadditivity](#) and the [characteristic-zero statement](#). The [additional inputs to WV](#) form a separate part of the article.

Exclude the automatic case where either right-hand term is $-\infty$, and put

$r = \kappa(F, K_F + \Delta_F) \geq 0$ and $b = \kappa(f | \Delta) \geq 0$. Below, X, W, Y denote allowed common models; (X_0, Δ_0) is the fixed smooth reference pair to which sections descend. The map $h : W \rightarrow Y$ retains the original base, while $p : W \rightarrow S$ is a separate map arising from periods.

Relative litaka reduction with a fixed reference

$$X \xrightarrow{g} W \xrightarrow{h} Y, \quad \dim W_y = r, \quad \kappa(G, K_G + \Delta_G) = 0$$

$$C = B(f, \Delta), \quad \kappa(Y, K_Y + C) \geq b$$

Construct the Hodge line and boundary from the Kodaira-zero generator
Lem. 2.6, p. 9; Prop. 2.7, pp. 10–17

Compare boundary orders and transfer sections

$$B(g, \Delta) \leq T \leq 1, \quad D_W = K_W + T$$

$$H^0(W, q(D_W + M)) \simeq H^0(X_0, q(K_{X_0} + \Delta_0))$$

Products and ratios are preserved, also over Y .

Relative comparison and multiplicities
Prop. 2.7(iv); Lem. 2.12
pp. 10, 17; (6.4)–(6.5), pp. 39–40

Adjoint positivity of the highest line
Thm. 3.1, pp. 17–18
Prop. 4.5; Lem. 4.7, pp. 25–31

Use the map to the original base

$$h : W \rightarrow Y$$

$$(D_W + M)|_{W_y} \text{ big}$$

$$B(h, T) \geq C$$

$$T_E = 1 \quad (\text{codim}_Y h(E) \geq 2)$$

Pass to the separate period base

$$p : W \rightarrow S, \quad S \text{ projective}$$

$$M = p^*P, \quad P \text{ nef}$$

$$K_S + a_0P \text{ big}, \quad a_0 > 0$$

No map between Y and S is assumed.

General-type addition and cancellation by one fixed section
Prop. 3.4, pp. 19–20; (6.6), p. 40

Combine the two inputs

$$\kappa(W, D_W + M) \geq r + \kappa(Y, K_Y + C)$$

Descent to the reference preserves ratios and image dimensions
Prop. 2.7; Lem. 2.1, pp. 5–6, 16–17; §6, p. 40

Return to the original pair

$$\kappa(X, K_X + \Delta) \geq r + b$$

Main theorem: the relative litaka reduction combines boundary control with Hodge-line positivity

Open figure ↗ TeX source ↓

1. Preserve exceptional coefficients and fix the original base

Resolve the base and flatten before resolving the source. Source primes mapping into base codimension at least two become exceptional over a fixed smooth reference. Taking the strict transform plus reduced exceptional boundary makes the difference of log canonical divisors effective and exceptional; pluricanonical sections, products, and ratios are preserved. Fix the resulting smooth base Y and set $C = B(f, \Delta)$. Further changes of W retain C and the coefficient-one condition, together with $\kappa(Y, K_Y + C) \geq b$.

Lemma 2.1 · Corollaries 2.3, 2.5 · Lemmas 2.4, 2.6 · pp. 5–9; §6 · p. 39

2. Replace the Kodaira-zero fiber by an actual Hodge line

The relative Iitaka factorization $X \xrightarrow{g} W \xrightarrow{h} Y$ has $\dim W_y = r$ and logarithmic Iitaka dimension zero on a very general g -fiber G . Take a root of a relative generating form s in a sufficiently divisible degree and use the highest Hodge part of its distinguished character. This does not assert that the entire highest Hodge space is a line. Coefficient-one boundary requires the mixed Hodge structure of an open fiber and passage to the unique nonzero pure weight grade. The extension theorem FF retains this identification at the boundary.

Lemmas 2.6, 2.8–2.10 · pp. 9–15 · FF, Theorem 1.1(ii), (iv) · p. 2

The boundary T compares the original form's allowed poles with its Hodge-extension order. In the notation of §2.6.3,

$$\alpha_D = \min_{E \mapsto D} \frac{l_E + \Delta_E}{a_E}, \quad \beta_D = \min_{E \mapsto D} \frac{l_E + 1 - a_E}{a_E}, \quad T_D = \alpha_D - \beta_D.$$

Here $a_E = \text{ord}_E(g^*D)$ and $l_E = m^{-1} \text{ord}_E(s \wedge g^*\omega^m)$, where ω is a local ordinary base volume form. The minima include every component of the prepared SNC source. This gives

$B(g, \Delta) \leq T \leq 1$ and, in sufficiently divisible degrees q ,

$$H^0(W, q(K_W + T + M)) \simeq H^0(X_0, q(K_{X_0} + \Delta_0)).$$

The untested primes mapping into base codimension at least two are exceptional over the reference. Descend first to X_0 , then extend across codimension two. Specifying this reference is the bridge from a generic formula to actual global section spaces.

Proposition 2.7 · (2.12), (2.14), (2.16) · pp. 10, 13–17

3. Recover the hypotheses for addition from relative section comparison

Proposition 2.7(iv) also identifies direct images over Y and preserves section ratios. The relative Iitaka image has dimension $r = \dim W_y$, so $K_W + T + M$ is big on very general h -fibers. Independently, the multiplication of weighted orders in Lemma 2.12 gives $B(h, T) \geq C$ and coefficient one on every prime of W mapping into codimension at least two in Y . Both are hypotheses required for addition.

Proposition 2.7(iv) · Lemma 2.12 · pp. 10, 17; (6.3)–(6.5) · pp. 39–40

4. Supply adjoint positivity on the period base

Theorem 3.1 starts with the highest Hodge line of a complex direct summand of a pure real-polarizable variation. After modification it supplies a smooth projective S with $M = p^*P$, P nef, and $K_S + a_0P$ big. The integral lattice belongs to the ambient variation; the chosen complex summand need not be rational.

The proof constructs a quotient from the full periods of rational adjoint factors. Compactification Vil and the Moishezon criterion BC provide a projective model. If D_S marks nonidentity local monodromy, including finite nonidentity monodromy, then $K_S + D_S$ is big. Lemma 4.6 uses BT and other inputs to assert $\nu(P|_{D_i}) < \nu(P)$ for each component. In Lemma 4.7, the available section count has degree $\nu(P)$ in the coefficient t , whereas the boundary loss has smaller degree. This makes $K_S + a_0P$ big. The rank of the chosen-line map is distinguished from the rank of the full period map.

Theorem 3.1 · pp. 17–18; Proposition 4.5 · Lemmas 4.6–4.7 · pp. 25–31 · Vil, Theorem 2.19 · p. 16 · BC, Theorem 1.1 / Corollary 1.3 · pp. 1–2 · BT, Theorem 1.1 · p. 2

5. Apply addition and return the sections to the original source

All hypotheses of Proposition 3.4 now hold. The section operation explained below gives

$$\kappa(W, K_W + T + M) \geq r + \kappa(Y, K_Y + C) \geq r + b.$$

Proposition 2.7 and exceptional descent return these systems to the original (X, Δ) . Ratios and image dimensions are preserved throughout, transferring the lower bound to the theorem's left side. If S is a point, M is rationally trivial and Proposition 3.2 applies directly.

Propositions 3.2, 3.4 · pp. 18–20; §6 · (6.6) · p. 40

Section 4 transfers this lower bound to a logarithmic base. The intermediate Theorem 3.1 also supplies WV's restricted Hodge comparison.

Ol, §6 · pp. 39–41 · WV, Theorem 3.8 / Proposition 3.9 · p. 21

3. The core of Proposition 3.4 — Cancel the ample twist with a fixed section

Write $D_W = K_W + T$ and $M = p^*P$. Start from bigness of $D_W + M$ on the fibers of h . The argument fixes rational coefficients and multiplies two systems; it makes no limiting-continuity assertion for κ .

Retain fiberwise bigness with a smaller coefficient

$$D_W = K_W + T, \quad M = p^*P, \quad 0 < c < 1$$

$$(D_W + cM)|_{W_y} \text{ big}, \quad A \text{ ample on } S$$

Add a general boundary Q_η and apply general-type-fiber addition
Prop. 3.2, p. 18; (3.5), p. 19; §5, pp. 31–38

Obtain enough ratios with a positive twist

$$E_+(\eta) = D_W + cM + \eta p^*A$$

$$\kappa(W, E_+(\eta)) \geq r + \kappa(Y, K_Y + C) \quad (\eta \in \mathbb{Q}_{>0})$$

Restrict to a general p -fiber, then use the big base twist
Lem. 3.3, pp. 18–19; (3.6)–(3.7), p. 20
[Fuj] Thm. 1.1, pp. 1–2

Produce one section with a negative twist

$$a > \max\{1, a_0\}, \quad H = K_S + aP - \lambda A \text{ big}$$

$$E_- = D_W + aM - \lambda p^*A, \quad 0 \neq e \in H^0(W, qE_-)$$

Set the coefficient of M to one and that of p^*A to zero
(3.8), p. 20

Fix rational coefficients and cancel exactly

$$\theta = \frac{1-c}{a-c}, \quad \eta = \frac{(1-c)\lambda}{a-1}$$

$$D_W + M = (1-\theta)E_+ + \theta E_-, \quad 0 < \theta < 1$$

Take sufficiently divisible n ; the common factor cancels in ratios
Prop. 3.4, following (3.8), p. 20

Preserve image dimension by fixed-section multiplication

$$H^0(W, n(1-\theta)E_+) \xrightarrow{\cdot e^{n\theta/q}} H^0(W, n(D_W + M))$$

$$\kappa(W, D_W + M) \geq r + \kappa(Y, K_Y + C)$$

Adjoint addition: combine a positively twisted system with one fixed negatively twisted section

Open figure ↗ TeX source ↓

1. Construct the positively twisted systems by general-type addition

Openness of the big cone gives $0 < c < 1$ with $D_W + cM$ still big on very general h -fibers. For an ample divisor A on S and any rational $\eta > 0$, the divisor $cP + \eta A$ is ample. A general rational member $Q_\eta \sim_{\mathbb{Q}} cM + \eta p^* A$ leaves $T + Q_\eta$ an SNC boundary, preserves old coefficient-one components, and retains the orbifold base lower bound. Proposition 3.2 gives

$$\kappa(W, D_W + cM + \eta p^* A) \geq r + \kappa(Y, K_Y + C).$$

To prove Proposition 3.2 over a nonprojective base, §5 slightly decreases only the horizontal boundary, obtaining klt general fibers. Lemma 5.2 compares these with a projective stable family of maximal variation. Theorem 8.1 and Corollary 8.3 of KP supply bigness of its relative log canonical divisor. Forms obtained on a finite cover descend through finite-group invariants; Lemma 5.4 retains their fiberwise image dimension. For the products with base sections, $C \leq B(h, T)$ controls poles above base divisors and $T_E = 1$ handles higher-codimension images. The original projective positivity theorem is thus applied to an auxiliary stable family.

Proposition 3.2 · p. 18; (3.5) · p. 19; Lemmas 5.2–5.5 · pp. 32–38 · KP, Theorem 8.1 / Corollary 8.3 · p. 40

2. Obtain one negatively twisted section by weak positivity

Restrict a nonzero section from the previous step to a very general p -fiber. The pulled-back twists disappear, giving relative nonvanishing for D_W . Choose $a > \max\{1, a_0\}$ and a small positive rational λ so that $H = K_S + aP - \lambda A$ is big. Lemma 3.3 uses Fuj to give a nonzero section of a positive multiple of

$$E_- = D_W + aM - \lambda p^* A = K_{W/S} + T + p^* H.$$

After flattening, the possible remaining poles are exceptional over the original smooth W , allowing descent to that fixed space.

Lemma 3.3 · pp. 18–19; (3.6)–(3.7) · p. 20 · Fuj, Theorem 1.1 · pp. 1–2

3. Cancel exactly and preserve ratios

Set

$$\theta = \frac{1-c}{a-c}, \quad \eta = \frac{(1-c)\lambda}{a-1}, \quad E_+ = D_W + cM + \eta p^* A.$$

Then $0 < \theta < 1$ and

$$D_W + M = (1 - \theta)E_+ + \theta E_-.$$

Fix $0 \neq e \in H^0(W, qE_-)$. For sufficiently divisible n , multiplication by $e^{n\theta/q}$ gives

$$H^0(W, n(1 - \theta)E_+) \hookrightarrow H^0(W, n(D_W + M)).$$

Away from the zeros of e , the common factor cancels from every ratio. The image dimension of the earlier systems is retained. Semiampneness of P is not required for this step.

Proposition 3.4 · (3.8) and the following multiplication · p. 20

Multiplication by a fixed section preserves image dimension, allowing the two positivity inputs in Section 2 to produce the main lower bound.

OI, Proposition 3.4 · §6 · pp. 19–20, 40

4. Proof of Corollary 6.2 — Compare with the logarithmic base

One needs both the inclusion of the prescribed boundary on a model and its transfer to the invariant $\kappa(f \mid D_X)$.

Support condition for reduced boundaries

$$\text{Supp}(f^* D_Y) \subseteq \text{Supp}(D_X)$$

$$f : X \rightarrow Y, \quad \Delta = D_X$$

Old and exceptional source components have coefficient one
[Lem. 6.1, p. 40](#)

Use infinite multiplicities on a neat model

$$q : Y' \rightarrow Y, \quad E_Y = (q^{-1} \text{Supp } D_Y)_{\text{red}}$$

$$B(f', \Delta') \geq E_Y$$

Pull back logarithmic forms and use the neat-model identity
[\(2.2\), p. 5; Lem. 6.1, p. 40](#)
[\[Cam\] Cor. 4.11, p. 49](#)

Pass to the invariant base term

$$\kappa(f \mid D_X) \geq \kappa(Y, K_Y + D_Y)$$

Substitute $\Delta = D_X$ into orbifold subadditivity
[Thm. 1.1, p. 3; Cor. 6.2, pp. 40–41](#)

Apply the main theorem to the logarithmic pair

$$\kappa(X, K_X + D_X) \geq \kappa(F, K_F + D_F) + \kappa(Y, K_Y + D_Y)$$

Logarithmic subadditivity: coefficient one yields an invariant base comparison, followed by the main theorem

[Open figure ↗](#) [TeX source ↓](#)

1. Pull back logarithmic systems on a neat model

In Lemma 6.1 take a neat model $f' : (X', \Delta') \rightarrow Y'$, with $q : Y' \rightarrow Y$, and put $E_Y = (q^{-1} \text{Supp } D_Y)_{\text{red}}$. Every source prime dominating a component of E_Y has coefficient one, so $B(f', \Delta') \geq E_Y$. Pullback of logarithmic volume forms preserves section ratios. The neat-model identity from [Cam](#) gives

$$\kappa(f \mid D_X) \geq \kappa(Y, K_Y + D_Y).$$

[Lemma 6.1 · p. 40; \(2.2\) · p. 5](#) [· Cam, Corollary 4.11 · p. 49](#)

2. Substitute into the main theorem and pass to open varieties

Apply Theorem 1.1 with $\Delta = D_X$. For a dominant morphism $U \rightarrow V$ of smooth quasi-projective varieties with geometrically connected general fiber, resolution of the graph and boundaries yields compatible compactifications satisfying Corollary 6.2. Thus $\bar{\kappa}(U) \geq \bar{\kappa}(U_v) + \bar{\kappa}(V)$ for very general v . Zero boundaries give ordinary subadditivity over $\mathbb{C}$.

Proof of Corollary 6.2 and the following discussion · pp. 40–41

This lower bound is used in WV’s parameter-field construction and in the corollary turning RA’s independent upper bound into equality.

WV, Theorem 2.1 · p. 6 · RA, Corollary 1.2 / §7.6 · pp. 2, 50

5. Proof of Corollary 6.3 — Geometric generic fibers and field extension

Section spaces on the geometric generic fiber transfer the inequality from very general complex fibers to algebraically closed fields of characteristic zero.

Descend characteristic-zero data to a finitely generated field

$K \subset k, \quad K/\mathbb{Q}$ finitely generated, $K \hookrightarrow \mathbb{C}$
 $f_K : (X_K, D_{X_K}) \longrightarrow (Y_K, D_{Y_K})$

Retain smoothness, boundary conditions, and connected fibers
Cor. 6.2; proof of Cor. 6.3, p. 41

Apply logarithmic subadditivity over the complex numbers

$$\kappa(X_{\mathbb{C}}, K + D_{X_{\mathbb{C}}}) \geq \kappa(F_{\mathbb{C}}, K + D_{F_{\mathbb{C}}}) + \kappa(Y_{\mathbb{C}}, K + D_{Y_{\mathbb{C}}})$$

Use base change, image dimension in each degree, and countably many exceptions
Proof of Cor. 6.3, p. 41

Pass from very general to geometric generic fibers

$$\kappa(F_{\text{very general}}, K + D) = \kappa(F_{\bar{\eta}}, K + D)$$

Preserve nonvanishing and complete-linear-system image dimensions
Cor. 6.3, p. 41

Return to the original field through section-space invariance

$$H^0(V, mL) \otimes_K K' \simeq H^0(V_{K'}, mL_{K'})$$

$$\kappa(X, K_X + D_X) \geq \kappa(F_{\bar{\eta}}, K + D_F) + \kappa(Y, K_Y + D_Y)$$

Set $D_X = D_Y = 0$ for ordinary subadditivity.

Passage to characteristic zero: degreewise base change and descent to a finitely generated field

[Open figure ↗](#) [TeX source ↓](#)

1. Compare sections and image dimensions degree by degree

For a complex projective family, apply cohomology and base change in every divisible degree, and choose opens on which the relative rational maps have constant fiberwise image dimension. Outside a countable union of exceptional sets, the very general and geometric generic fibers have the same Iitaka dimension. For a field extension $K \subseteq K'$,

$$H^0(V, \mathcal{O}_V(mL)) \otimes_K K' \simeq H^0(V_{K'}, \mathcal{O}_{V_{K'}}(mL_{K'}))$$

preserves nonvanishing and the image dimension of the complete linear system.

[Proof of Corollary 6.3 · p. 41](#)

2. Descend to a finitely generated field and apply the complex inequality

Descend f , its projective embeddings, and boundaries to a subfield $K \subset k$ finitely generated over $\mathbb{Q}$, and choose $K \hookrightarrow \mathbb{C}$. Smoothness, geometric integrality, and boundary conditions descend, as does $f_*\mathcal{O}_X = \mathcal{O}_Y$ by faithful flatness. Apply the complex logarithmic inequality and identify all three Iitaka dimensions with those over k by field-extension invariance.

Proof of Corollary 6.3 · p. 41

Ordinary and logarithmic subadditivity are now available over algebraically closed fields of characteristic zero. PH's ordinary subadditivity is a separate proof.

Ol, Corollary 6.3 · p. 41 · PH, Theorem 1.1 · p. 2

6. Additional results supplied to whole-fiber variation

Section 7 is an additional route following the main theorem. Lemma 7.1 concerns a smooth geometrically integral projective pair (J, D) over a field K of characteristic zero, with reduced SNC D and $\kappa(J_{\bar{K}}, K_{J_{\bar{K}}} + D_{\bar{K}}) = 0$. Taking a root of a form in the least nonvanishing degree p gives a cover with an appropriate pole boundary $D_{\tilde{N}}$ and

$$H^0(\tilde{N}, m(K_{\tilde{N}} + D_{\tilde{N}})) = K\Omega^m \quad (m \geq 1).$$

Corollary 7.2 realizes this **entire highest piece** in a pure integral variation and identifies the normalization of its rational extension with M_W . This strengthens the character-line conclusion of Proposition 2.7 for general rational boundaries, but requires reducedness. In Remark 7.3, five points of coefficient $2/5$ on $\mathbb{P}^1$ give a root cover with six-dimensional highest space.

§7 · Lemma 7.1 · Corollary 7.2 · Remark 7.3 · pp. 43–46

Proposition 9.1 of WV uses these results together with Proposition 2.7 and Lemma 7.4 for an additional relative Iitaka construction. The separately checked inputs to its main route are the lower bound from Corollary 6.2 and adjoint positivity from Theorem 3.1. Theorem 6.28 of BFMT used to identify the moduli divisor in §7, the full valuation comparison of Lemma 7.4, and the complete arguments of WV §§8–9 remain unchecked here. BFMT's separate semiamplessness theorem should not be substituted for this identification.

WV, Theorem 2.1 · p. 6; Theorem 3.8 / Proposition 3.9 · p. 21; Proposition 9.1 · pp. 69–70 ·
Ol, Corollary 7.2 · pp. 45–46

Lemma 7.4 compares the minimum over actual source components with the lct allowing valuations on higher models, on the same SNC preparation. WV Proposition 9.1 uses this statement to normalize the initial boundary so that section comparison and the entire highest line are handled on the same preparation. The input statement and this use were compared; the unverified scope above concerns the full valuation and model-change arguments.

OI, Lemma 7.4 · pp. 46–47 · WV, proof of Proposition 9.1 · p. 70

7. Which papers supply which steps

Input

FF Theorem 1.1(ii), (iv), p. 2

Use and role in this manuscript

§2.6.4–5, pp. 13–15: canonical extension of the logarithmic highest part and passage to a pure weight grade

Scope checked here

Statement and application compared; the full semistable construction is unchecked

Input

Vil Theorem 2.19, p. 16

Use and role in this manuscript

Proposition 4.5, pp. 25–26: compactification of the class-C period quotient

Scope checked here

Statement and application compared

Input

BC Theorem 1.1 / Corollary 1.3, pp. 1–2

Use and role in this manuscript

Proposition 4.5 and proof of Theorem 3.1, pp. 26, 30–31: projective model and $K_S + D_S$ big

Scope checked here

Statements and stated application hypotheses compared

Input

BT Theorem 1.1, p. 2

Use and role in this manuscript

Lemma 4.6, p. 27: expected dimension on the Zariski closure of a line-map fiber

Scope checked here

Statement and application compared; the full rank-loss argument is unchecked

Input

KP Corollaries 6.19, 7.3, pp. 28–29; Theorem 8.1 / Corollary 8.3, p. 40

Use and role in this manuscript

Lemmas 5.2, 5.4, pp. 34, 36: projective stable family and bigness of its relative log canonical divisor

Scope checked here

Statements, maximal variation, and klt general-fiber conditions compared

Input

Fuj Theorem 1.1, pp. 1–2

Use and role in this manuscript

Lemma 3.3, pp. 18–19: nonvanishing after a big base twist

Scope checked here

Statement and class-C lc-pair/projective-base hypotheses compared

Input

Cam Corollary 4.11, p. 49

Use and role in this manuscript

(2.2), p. 5; Lemma 6.1, p. 40: invariant base term on a neat model

Scope checked here

Statement and application compared; the full neat-model theory is unchecked

Within catalogue 033, [OI](#) Corollary 6.2 supplies [WV](#) Theorem 2.1, and [OI](#) Theorem 3.1 supplies [WV](#) Theorem 3.8. In [RA](#), Corollary 6.2 is used only for the additivity conclusion of Corollary 1.2, not for the upper inequality in Theorem 1.1. [PH](#) is presented as an alternative ordinary proof. Connections to [LA](#) and the Kähler manuscripts in catalogue 034 should remain attached to individual results; the path [OI](#)→[LA](#)→[Schnell](#) is not a direct [OI](#)→[Schnell](#) dependency.

[WV](#) · pp. 6, 21, 69–70 · [RA](#) · pp. 2, 50 · [OI](#) · pp. 5, 41

8. Return to the sources

The manuscript is by OpenAI, dated September 26, 2026, with 55 pages including references. The reference commit is `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Start with the assembly in §6, pp. 39–40, then return to the fixed-reference section comparison in Proposition 2.7, pp. 10–17, and cancellation in Proposition 3.4, pp. 19–20. The two positivity inputs are proved in §4, pp. 20–31, and §5, pp. 31–38.

The passages read here cover the model convention, relative Iitaka reduction, order comparison and descent in §2; effectivity and cancellation in §3; the period quotient, the application portion of rank loss, and boundary section counting in §4; the stable-family statement and main construction steps, finite descent and pole tests in §5; and the main proof and Corollaries 6.2–6.3 in §6. For §7, the statements and proof routes of Lemma 7.1 and Corollary 7.2 were read. Statements and applications of the external inputs in the table were compared.

The complete rational-adjoint construction, all of Lemma 4.6 including finite monodromy, technical connections in semistable reduction and canonical extension, the full parameter-space construction for stable-family comparison, Lemma 7.4 and Appendix A, the complete proofs of the core consequences in Corollaries 6.4–6.5, and the external proofs themselves have not been independently verified. This draft explains the manuscript's claims and argument; it does not certify the correctness of the full proof. Unrecorded relations between papers remain uninvestigated.

[Complete manuscript](#)

Logarithmic Kodaira dimension and whole-fiber variation

From a field detected by logarithmic Iitaka systems to a field of definition of the whole fiber

For a projective fibration over a base of nonnegative logarithmic Kodaira dimension, the manuscript adds whole-fiber variation to the Kodaira-dimension lower bound. Its argument constructs a parameter field from logarithmic sections, establishes birational constancy of a minimal-index root cover, fixes a finite normalization of the relative Iitaka base using markings, and descends a finite cover.

Manuscript author: OpenAI · 2026-09-26 · 75 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

These are the manuscript's claims, with its numbering, hypotheses, and conclusions retained. Fix $\Omega = \overline{\mathbb{C}(V)}$ and let F be the geometric generic fiber. The variation is

$$\mathrm{Var}(f) = \min_{\substack{\mathbb{C} \subseteq L \subseteq \Omega \\ L \text{ algebraically closed}}} \{\mathrm{trdeg}_{\mathbb{C}} L : F \text{ is birational to } (F_L)_{\Omega} \text{ for some } F_L/L\}.$$

It measures the function field of the entire F . Variation of a period image, a canonical model, or the relative Iitaka base alone is a different invariant.

[Equation \(1.1\) · p.2](#)

Theorem 1.1 — Logarithmic variation

Let $f : U \rightarrow V$ be a projective surjection with connected fibers between smooth connected complex quasi-projective varieties. If $\bar{\kappa}(V) \geq 0$, then

$$\bar{\kappa}(U) \geq \kappa(F) + \max\{\bar{\kappa}(V), \mathrm{Var}(f)\}.$$

The morphism need not be smooth. No abundance or good-minimal-model hypothesis is imposed on F . When $\kappa(F) = -\infty$, the assertion uses the manuscript's convention $(-\infty) + a = -\infty$.

[Theorem 1.1 · p.2](#)

Corollary 1.2 — Projective Iitaka–Viehweg inequality

Let $f : X \rightarrow Y$ be a surjection with connected fibers between smooth connected complex projective varieties, and assume $\kappa(Y) \geq 0$. Then

$$\kappa(X) \geq \kappa(F) + \max\{\kappa(Y), \text{Var}(f)\}.$$

[Corollary 1.2 · p.2](#)

Corollary 1.3 — Variation and birational isotriviality

Let $f : U \rightarrow V$ be a **smooth projective** surjection with connected fibers between smooth connected complex quasi-projective varieties. Assume every closed fiber is non-uniruled. Then

$$\begin{aligned} \bar{\kappa}(V) = -\infty &\implies \text{Var}(f) < \dim V, \\ \bar{\kappa}(V) \geq 0 &\implies \text{Var}(f) \leq \bar{\kappa}(V). \end{aligned}$$

If V is Campana-special, $\text{Var}(f) = 0$. In particular, $\bar{\kappa}(V) = 0$ implies birational isotriviality. Here isotriviality means variation zero; no trivialization of the family by isomorphisms is asserted. This corollary additionally uses [LA](#) and [Taji](#) and is not an input to [Theorem 1.1](#).

[Corollary 1.3 and its proof · p.5](#)

References on the arrows

Unprefixed result, equation, and section numbers refer to [WV](#). GitHub links open a fixed-commit manuscript preview; consult the PDF pages given in the labels.

- [OI](#): Orbifold and logarithmic Iitaka subadditivity, 2026-09-26. Subadditivity and adjoint positivity.
- [Fuj](#): Fujino, Notes on the weak positivity theorems, 2015-06-30, version 0.54.
- [Del](#): Deligne, Théorie de Hodge II, 1971. Finite characters and invariant cycles.
- [DHP](#): Demailly–Hacon–Păun, Extension theorems, non-vanishing and the existence of good minimal models, arXiv v2.
- [PT](#): Păun–Takayama, Positivity of twisted relative pluricanonical bundles and their direct images, arXiv v1.

- **BCHM**: Birkar–Cascini–Hacon–McKernan, Existence of minimal models for varieties of log general type, the 81-page arXiv v2.
- **KP**: Kovács–Patakfalvi, Projectivity of the moduli space of stable log-varieties and subadditivity of log-Kodaira dimension, the 62-page author manuscript.
- **Han**: Hanamura, Structure of birational automorphism groups, I: non-uniruled varieties, 1988. The original theorem statements were not compared in this reading.
- **SGA1**: Revêtements étales et groupe fondamental, 2003 re-edition. The original purity statement was not compared in this reading.
- **Lan**: Landesman, Invariance of the tame fundamental group under base change between algebraically closed fields, arXiv v3.
- **BDPP**: Boucksom–Demailly–Păun–Peternell, The pseudo-effective cone of a compact Kähler manifold and varieties of negative Kodaira dimension, arXiv v1.
- **LA**: Log abundance in characteristic zero, 2026-09-24. **Taj**: Taji, Birational geometry of smooth families of varieties admitting good minimal models, arXiv v4.

2. Overview of Theorem 1.1

Proof route. The main theorem has three stages: construct the parameter field, establish constancy of the Kodaira-zero fibre, and descend the original whole fibre. The empty-boundary corollary and the smooth-family consequence using LA and Taji are treated separately after the main proof.

For $d = \kappa(F) \geq 0$, make the field $b = \overline{\mathbb{C}(T_0)}$ detected by section systems into a birational definition field of the original whole geometric generic fiber F . The Kodaira-zero relative Iitaka fiber J is distinct from F .

Compare the logarithmic litaka system with the original base

$$d = \kappa(F) \geq 0, \quad \bar{\kappa}(V) \geq 0, \quad X' \longrightarrow W \longrightarrow Z$$

Use the logarithmic lower bound and proportional forms
[\[OI\] Cor. 6.2, pp. 40–41](#); [\[WV\] Prop. 2.10, pp. 12–14](#)

Embed the parameter field into the original base field

$$\mathbb{C}(T_0) \subseteq \mathbb{C}(Y), \quad \bar{\kappa}(U) = d + \dim T_0$$

Restricted adjoint comparison, contraction of poles and equivariant descent
[\[OI\] Thm. 3.1, pp. 17–18](#); [\[WV\] Prop. 3.12, p. 23](#)
[\[WV\] Cor. 3.16, p. 25](#); [Thm. 4.1 / Prop. 4.10, pp. 26, 42–43](#)

First birational constancy along the litaka fibers

$M|_{W_z} \sim_{\mathbb{Q}} 0 \implies$ flat top line with finite character
 From constancy of the root cover to constancy of J

Retain coordinates, actions and covers beyond constancy of J
[\[WV\] Prop. 6.2, pp. 47–51](#); [Lem. 7.1 / Prop. 7.2, pp. 52–56](#)

Fix the litaka-base coordinates and the whole function field

$b = \overline{\mathbb{C}(T_0)}$, $C_* \rightarrow P_0$, $V' \rightarrow P_0$
 Markings, constant cocycles, and no ramification on moving divisors

Descend the cover and action; identify the original geometric generic fiber
[\[WV\] Prop. 7.3 / §7.4, pp. 57–59](#)

Insert whole-fiber variation into the lower bound

$$\mathrm{Var}(f) \leq \dim T_0, \quad \bar{\kappa}(U) \geq d + \max\{\bar{\kappa}(V), \mathrm{Var}(f)\}$$

Overview linking the parameter field, first constancy, markings and descent of finite covers

[Open figure ↗](#) [TeX source ↓](#)

1. Obtain the candidate field from section systems

OI's logarithmic lower bound yields $\mathbb{C}(T_0) \subseteq \mathbb{C}(Y)$ and $\bar{\kappa}(U) = d + \dim T_0$. The next section develops this construction.

Proposition 2.10, Remark 2.11 · pp.12–14 · Ol, Corollary 6.2 · pp.40–41

2. Turn Hodge-line triviality into birational constancy

Apply Ol's adjoint comparison to the restricted root-cover variation to make its entire top line flat with finite character. Contraction of poles and local flows make J constant; article Section 4 develops this step.

Propositions 3.9, 3.12, Corollary 3.16, Proposition 4.10 · pp.21–25, 42–43 · Ol, Theorem 3.1 · pp.17–18

3. Return to the whole fiber while retaining coordinates and actions

Markings fix litaka-base coordinates and a finite normalization. Constant cocycles and inertia at moving divisors allow descent of covers and actions and identification of the original function field. Article Section 5 obtains $\text{Var}(f) \leq \dim T_0$.

Proposition 6.2, Propositions 7.2–7.3, §7.4 · pp.47–59

Combine this bound with the section-system dimension identity to obtain Theorem 1.1. Corollary 1.2 sets the boundaries to zero; Corollary 1.3 is a separate consequence using LA and Taji. The diagrams for the [parameter field](#), [first constancy](#), and [whole-fibre descent](#) expand these three stages.

3. Theorem 1.1: constructing the parameter field

Assume $d = \kappa(F) \geq 0$. The first task is to compare the absolute logarithmic litaka base Z and the original base Y inside the same function field, producing a subfield of $\mathbb{C}(Y)$ of transcendence degree $\bar{\kappa}(U) - d$. No constancy assertion is used at this stage.

01 Logarithmic lower bound and a base pluriform

$$d = \kappa(F) \geq 0, \quad \bar{\kappa}(U) \geq d + \bar{\kappa}(V),$$

$$0 \neq \xi \in H^0(Y, m(K_Y + D_Y)).$$

Retain the reduced inverse boundary; take the relative algebraic closure of the compositum.

[\[OI\] Cor. 6.2, pp.40–41](#) [Lemma 2.6; Prop. 2.7, pp.8–10](#)

02 Combine the absolute litaka map with the original fibration

$$X' \xrightarrow{x} W, \quad W \xrightarrow{g} Y, \quad W \xrightarrow{h} Z, \quad q = hx.$$

$\dim Z = \bar{\kappa}(U)$, $W \rightarrow Y \times Z$ is generically finite onto its image.

Contract $g^*\xi$ with $p_1 = \dim Y - \dim W_z$ base directions.
Subadditivity for x_z forces both nonnegative terms to vanish.

[Lemmas 2.8–2.9, pp.10–12](#)

03 Force proportionality on litaka fibers

$$\kappa(J) = \kappa(J, K_J + D'|_J) = 0,$$

$$\kappa(W_z, K_{W_z} + D_W^0|_{W_z}) = 0, \quad D_W^0 = (g^*D_Y)_{\text{red}}.$$

Constant ratios of minors give the rank of the image family.
Relative algebraic closedness in $\mathbb{C}(Y) \subseteq \mathbb{C}(S) \subseteq \mathbb{C}(X)$ gives $\mathbb{C}(S) = \mathbb{C}(Y)$.

[Prop. 2.10, \(2.10\)–\(2.12\), pp.12–14](#)

04 Recover a subfield of the original base from the image family

$$\mathbb{C}(T_0) \subseteq \mathbb{C}(Y), \quad \bar{\kappa}(U) = d + \dim T_0.$$

Remaining goal: F descends to $b = \overline{\mathbb{C}(T_0)}$.

Logarithmic subadditivity and the family of images produce an embedded parameter field

[Open figure ↗](#) [TeX source ↓](#)

1. Combine the two maps while retaining coefficient-one boundary

Choose SNC compactifications $(X, D_X) \rightarrow (Y, D_Y)$ with $f^{-1}(V) = U$. Applying [OI Corollary 6.2](#) gives $\bar{\kappa}(U) \geq d + \bar{\kappa}(V)$. Resolve the absolute litaka map of $K_X + D_X$ as $q : X' \rightarrow Z$. The relative algebraic closure of $\mathbb{C}(Y)\mathbb{C}(Z)$ in $\mathbb{C}(X)$ defines $X' \xrightarrow{x} W \xrightarrow{h} Z$ and $g : W \rightarrow Y$. The map $W \rightarrow Y \times Z$ is generically finite onto its image, and the relevant generic fibers are geometrically integral.

[Lemma 2.6, Proposition 2.7 · pp.8–10](#) · [OI, Corollary 6.2 · pp.40–41](#)

2. Contract logarithmic forms to compute the rank of the image family

Fix $0 \neq \xi \in H^0(Y, m(K_Y + D_Y))$. Set $r = \dim W_z$ and $p_1 = \dim Y - r$. Contracting $g^*\xi$ against p_1 directions from Z produces a nonzero logarithmic pluricanonical form on W_z . Since X'_z has logarithmic Kodaira dimension zero, subadditivity for x_z gives

$$\kappa(W_z, K_{W_z} + D_W^0|_{W_z}) = 0, \quad \kappa(J) = \kappa(J, K_J + D'|_J) = 0, \quad D_W^0 = (g^*D_Y)_{\text{red}}.$$

Here J is the geometric generic fiber of x . The contracted sections in a fixed degree are proportional. Consequently the ratios of maximal minors of the normal deformation map $T_z Z \rightarrow N_{g(W_z)/Y, y}$ do not depend on y . Its constant kernel gives rank p_1 for the differential of the Hilbert parameter map.

Lemmas 2.8–2.9, proof of Proposition 2.10 · pp.10–13

3. Embed the parameter field in the original base field

Take the relative algebraic closure in $\mathbb{C}(Z)$ of the parameter field of the image family $g(W_z)$, and call its smooth model T_0 . The resolved universal image family S maps generically finitely to Y . In the tower

$$\mathbb{C}(Y) \subseteq \mathbb{C}(S) \subseteq \mathbb{C}(W) \subseteq \mathbb{C}(X),$$

the first extension is algebraic, while $\mathbb{C}(Y)$ is relatively algebraically closed in $\mathbb{C}(X)$. Thus $\mathbb{C}(S) = \mathbb{C}(Y)$, an identification stronger than a dimension count. Together with $\dim(W/Y) = d$, it gives

$$\mathbb{C}(T_0) \subseteq \mathbb{C}(Y), \quad \bar{\kappa}(U) = \dim Z = d + \dim T_0.$$

It remains to descend the entire fiber to $b = \overline{\mathbb{C}(T_0)} \subseteq \Omega$.

Proposition 2.10, Remark 2.11 · pp.12–14

Fix the resulting $b = \overline{\mathbb{C}(T_0)}$ as the candidate definition field. Constancy of J , followed by markings and descent of finite covers, must still recover the original F .

WV, §§3–7 · pp. 14–59

4. Theorem 1.1: the root cover and the first constancy statement

The least ordinary pluricanonical index makes the **entire** highest Hodge piece rank one, rather than just an eigenspace. Manuscript Section 4 supplies the analytic passage from its flatness to birational constancy.

01 Take the root cover at the least index

$$p = \min\{m > 0 : H^0(J, mK_J) \neq 0\}, \quad \eta^{\otimes p} = \pi^* \omega_J.$$

$$h^0(\tilde{J}, mK_{\tilde{J}}) = 1 \ (m > 0) : \quad \text{the entire highest Hodge piece is a line.}$$

The Hodge integrability threshold gives the parabolic order.
Multiply by the relative form and transfer ratios to the fixed $X_{*,z}$.

Lemma 3.1; Props. 3.3, 3.5, 3.7, pp.14–21

02 Compare section systems using actual relative orders

$$B_P = 1 - \lambda_P + t_P, \quad M \sim_{\mathbb{Q}} \sum_P (\lambda_P - 1)P, \quad B \geq D_W^0,$$

$$\kappa(W_z, K_{W_z} + B_z + M_z) \leq 0.$$

Apply comparison to the restricted integral variation. Weak positivity gives
 $\kappa(D + cM_z) \geq \dim R$, contradicting the upper bound if $\dim R > 0$.

[OI] Thm. 3.1, pp.17–18 [Fuj] Thm. 1.1, pp.1–2

Props. 3.9, 3.12; Lemma 3.11, pp.21–24

03 Interpolate the adjoint comparison on the restriction

$$D = K_{W_z} + B_z \sim_{\mathbb{Q}} D_{\text{eff}}, \quad M_z \sim_{\mathbb{Q}} a^* A,$$

$$D + M_z = c^{-1}(D + cM_z) + (1 - c^{-1})D,$$

$$\dim R = 0 \implies M_z \sim_{\mathbb{Q}} 0, \quad \kappa(D) = 0.$$

Vanishing semipositive curvature on curves gives a line subsystem; apply finite character on the restriction.

Cors. 3.13–3.16, pp.24–25 [Del] Cor. 4.2.8(iii)(b), PDF p.44

04 Obtain a flat entire top line with finite character

$$\deg M_z|_C = 0 \implies \text{constant projective top line; finite character.}$$

Extension, Bergman weights, and minimization with a prescribed leading term make poles fixed. Contract on a big adjoint model, spread flows, and take the equivariant quotient.

[DHP] Thm. 4.1, pp.20–21 [PT] Thm. 4.2.7, p.36

[BCHM] Thm. 1.2(2), p.5 Thm. 4.1; Prop. 4.10, pp.26–43

05 Contract poles to obtain the first constancy statement

$$\sigma_S(K_D) > 0 \text{ for pole divisors, } K_D + \varepsilon H \text{ big klt.}$$

J is geometrically birationally constant along h -fibers.

The minimal root cover, restricted Hodge-line triviality, and analytic contraction give the first constancy statement

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1. Relate actual relative-form orders to the Hodge line

The connected cyclic cover obtained from the least $p > 0$ and $0 \neq \omega_J \in H^0(J, pK_J)$ has a resolution $\tilde{J}$ with $h^0(\tilde{J}, mK_{\tilde{J}}) = 1$ for every $m > 0$. Its root form spans the entire top-form space. For a base divisor P , distinguish the minimum t_P on actual source components from the threshold λ_P on all higher models:

$$t_P = \min_{Q \subset \tilde{X}', Q \rightarrow P} \frac{r_Q + d_Q}{m_Q}, \quad \lambda_P = \inf_{Q \rightarrow P} \frac{1 + r_Q}{m_Q}, \quad B_P = 1 - \lambda_P + t_P.$$

Here $m_Q = \text{ord}_Q(x^*P)$, $r_Q = p^{-1} \text{ord}_Q(\omega_J)$ is measured in the actual relative canonical bundle, and $d_Q = \text{coeff}_Q(D')$. The integrability exponent of the Hodge norm identifies the parabolic order of its line M as $\lambda_P - 1$. Thus $T = \sum t_P P \sim_{\mathbb{Q}} B + M$, $0 \leq B \leq 1$, and $B \geq D_W^0$.

Lemma 3.1, Propositions 3.3 and 3.5 · pp.14–19

2. Apply adjoint comparison anew to the restricted variation

Multiplication by the relative generator transfers sections to the fixed $X_{*,z}$ while preserving their ratios, giving

$$\kappa(W_z, K_{W_z} + B|_{W_z} + M|_{W_z}) \leq 0.$$

Meanwhile $D = K_{W_z} + B|_{W_z}$ is rationally effective. Apply [OI Theorem 3.1](#) to the **restricted integral variation** to obtain $M|_{W_z} \sim_{\mathbb{Q}} a^*A$ with $K_R + cA$ big. Weak positivity from [Fuj](#) and section transfer give $\kappa(D + cM) \geq \dim R$. For $c \geq 1$,

$$D + M = c^{-1}(D + cM) + (1 - c^{-1})D$$

transfers that lower bound to $D + M$, forcing R to be a point. Hence $M|_{W_z} \sim_{\mathbb{Q}} 0$ and $\kappa(W_z, D) = 0$. Curvature on test curves and [Del's](#) finite-character theorem, applied on the restriction, give a flat entire top line with finite monodromy.

[Propositions 3.7, 3.9 and 3.12](#), [Lemma 3.11](#), [Corollaries 3.13–3.16](#) · pp.20–25 · [OI, Theorem 3.1](#) · pp.17–18 · [Fuj, Theorem 1.1](#) · pp.1–2 · [Del, Corollary 4.2.8\(iii\)\(b\)](#) · printed p.47 / PDF p.44

3. Contract poles of the lifted vector field

Over a curve, kill the finite character and lift the invariant class to a global form Θ . Its kernel defines a rational vector field with $d\pi(v) = \partial_t$ and $\iota_v \Theta = 0$. Poles may remain along the zero divisor of the relative top form. [Lemma 4.4](#) uses [DHP](#) extension, while [Lemmas 4.6–4.7](#) use [PT](#) relative Bergman weights. These lead to positive limiting fixed order $\sigma_S(K_D) > 0$ at every pole divisor S . The weight is constructed before the final fiber is selected, and the regularization, degree, and cutoff limits have a prescribed order.

[Sections 4.1–4.5](#) · pp.26–40 · [DHP, Theorem 4.1](#) · pp.20–21 · [PT, Theorem 4.2.7 and Remark 4.3.1](#) · pp.36–37

Choose one small $\varepsilon > 0$ for all the finitely many poles. The big klt adjoint model of $K_D + \varepsilon H$, supplied by BCHM, contracts their positive fixed orders. Normality extends the resulting regular vector field across codimension two. Local flows and algebraic spreading of graphs give birational constancy after a finite field extension. Applying this equivariantly to the root cover and taking invariant function fields makes J constant in the h -fiber directions. Descending the original whole F still requires the next step.

Proof of Theorem 4.1, Corollary 4.9, Proposition 4.10 · pp.40–43 · BCHM, Theorem 1.2(2) · p.5

This fixes the Kodaira-zero fiber J . The next descent retains litaka-base coordinates and group actions to fix the entire geometric generic fiber.

WV, §6 introduction · §7.4 · pp. 46, 58–59

5. Theorem 1.1: descend the whole fiber with its litaka-base coordinates

Put $b = \overline{\mathbb{C}(T_0)}$ and denote the geometric generic fibers of W, S, Z over T_0 by H_0, I, P_0 . Write $p_I : H_0 \rightarrow I$ and $a_0 : H_0 \rightarrow P_0$ for the induced maps. The map $H_0 \rightarrow I \times_b P_0$ is generically finite and $\dim P_0 = d$. Constancy of J must now be made compatible with the coordinates of P_0 .

01 Add markings recognized by their coefficients

$$H_0 \longrightarrow I \times_b P_0, \quad \dim P_0 = d, \quad b = \overline{\mathbb{C}(T_0)},$$

$$(x_0 = 0), (x_j = 0), (x_0 + x_j = 0) : \quad d + \text{Var}(\text{marked model}) \leq \kappa(K_{H_0} + \Delta) \leq d.$$

Force marked-model variation zero; sum markings fix the remaining scalars.
Use the first constancy statement to choose a reference fiber J_0/k .

[\[KP\] Thm. 9.9, p.50](#) [Props. 5.3, 6.2, pp.45–51](#)

02 Fix the map, finite normalization, and old-boundary images

$$x_j/x_0 = c_j f_j, \quad f_j \in b(C_*), \quad c_j \in b,$$

$$V' \longrightarrow P_0 \text{ finite}, \quad k = b(V'), \quad K = k(I), \quad J_0/k.$$

Average the abelian part of the birational cocycle. Fixed old-boundary images
and equality in the order test produce an actual Q after every finite parameter extension.

[Lemmas 6.4, 7.1, pp.51–54](#) [\[Han\] Thms. 2.1–2.2](#)

Original [Han] statements not compared.

03 Combine a constant cocycle with a multiplicity-one minimizer

$$\tilde{K}/K \text{ Galois}, \quad \rho_g = z_g s_g, \quad (z_g, g|_{k'}) \text{ faithful},$$

$$P \text{ moving} : \quad B_P = 0, \quad m_Q = 1, \quad (1 + r_Q)/m_Q = \lambda_P.$$

The constant family's special fiber uniquely minimizes the threshold.
Equal inertia and faithful cocycle pairs force unramifiedness.

[Prop. 7.2, \(7.12\)–\(7.16\), pp.55–56](#)

04 Remove ramification at moving divisors

$$m_{Q'} = e_Q m_Q / e, \quad e_Q = e, \quad I_{Q'} = I_{P'} = 1.$$

Remove fixed branching and apply purity. Descend the finite étale cover, action,
and coefficient field together, then take invariants retaining the k -coordinates.

[\[Lan\] Thm. 1.1, p.2](#) [Prop. 7.3; Section 7.4, pp.57–59](#)

[\[SGA1\] X, Thm. 3.1](#) Original statement not compared.

05 Descend the function field of the whole fiber

$$k = b(V') \subseteq F_b = E_1(J_0)^{G_1},$$

$$\text{Frac}(\Omega \otimes_b F_b) \simeq \text{Frac}(\Omega \otimes_{\mathbb{C}(Y)} \mathbb{C}(X)),$$

$$\text{Var}(f) \leq \dim T_0 \implies \bar{\kappa}(U) \geq d + \max\{\bar{\kappa}(V), \text{Var}(f)\}.$$

Markings fix a finite normalization, and unramifiedness at moving divisors permits descent of the entire function field

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1. Recover the finite normalization using coordinate and sum markings

For an embedding of P_0 with coordinates $x_0, \dots, x_N$, label the pullbacks of $x_0 = 0$, $x_j = 0$, and $x_0 + x_j = 0$ by distinct small coefficients. Slightly decrease the old boundary to make the generic pair klt, and add further general markings to make it big. The added line bundles restrict trivially to a_0 -fibers, preserving the upper bound $\kappa(H_0, K_{H_0} + \Delta) \leq d$. Proposition 5.3, using KP, gives the lower bound $d + \text{Var}(\text{marked model})$. The marked model is therefore constant after a finite parameter extension. Let C_* be the underlying normal projective variety of this model pair over b .

Proposition 5.3, Proposition 6.2 · pp.45–50 · KP, Theorem 9.9 · p.50

Coordinate divisors determine $x_j/x_0 = c_j f_j$ with $f_j \in b(C_*)$; the sum divisors force $c_j \in b$. The morphism $C_* \rightarrow P_0$ itself descends, fixing the normalization $V' \rightarrow P_0$ in its function field. Images of positive old-boundary components become fixed b -divisors as well. With $k = b(V')$ and $K = k(I)$, the preceding constancy statement supplies a reference variety J_0/k . The markings are auxiliary identifiers and are discarded from the final fiber.

Proposition 6.2, especially (6.12) · pp.47–51

2. Remove the nonconstant cocycle and the inertia at moving divisors

Lemma 7.1 uses Han's structure theorem for birational automorphism groups. The abelian identity component and the unique divisibility of $A_1(\overline{K})/A_1(\overline{k})$ allow averaging over a finite quotient to remove the nonconstant cocycle as a coboundary. The resulting finite Galois splitting has a constant cocycle with faithful pairs $(z_g, g|_{k'})$.

Lemma 7.1 · pp.52–54

Comparison with the original Han statements remains pending.

At a moving divisor P , the old boundary coefficient is $B_P = 0$. Equality in the order inequalities produces an **actual source component** Q with $m_Q = 1$ attaining λ_P . On the split constant family the special fiber is the unique minimizing divisor. The identities

$$m_{Q'} = \frac{e_Q m_Q}{e}, \quad \frac{1 + r_{Q'}}{m_{Q'}} = e \frac{1 + r_Q}{m_Q} - (e - 1)$$

give $e_Q = e$. Total and base inertia coincide, and faithfulness of the constant cocycle makes inertia trivial. Repeating the argument after every finite parameter extension is needed to control all moving divisors over the algebraic closure.

Corollary 3.4 · p.18, Lemma 6.4 · pp.51–52, Proposition 7.2 · pp.55–56

3. Descend the cover, its action, and its coefficient field together

Remove the fixed branch divisors and singular locus to obtain $\mathcal{O} \subset V'$. The manuscript invokes purity to make the splitting cover finite étale over $\mathcal{O}$. In characteristic zero, Lan's base-change invariance descends covers and their morphisms, retaining both the group action and the map carrying k' . For the field E_1 of a connected component and its stabilizer G_1 , set

$$F_b = E_1(J_0)^{G_1}, \quad k = b(V') \subseteq F_b.$$

Commutation of finite-group invariants with field extension and identification of the original field tower give

$$\mathrm{Frac}(\Omega \otimes_b F_b) \simeq \mathrm{Frac}(\Omega \otimes_{\mathbb{C}(Y)} \mathbb{C}(X)).$$

The right side is the function field of the entire original geometric generic fiber. Hence $\mathrm{Var}(f) \leq \mathrm{trdeg}_{\mathbb{C}} b = \dim T_0$. Combine this with the dimension identity and subadditivity from article Section 3 to obtain Theorem 1.1.

Proposition 7.3 and Section 7.4 · pp.57–59 · Lan, Theorem 1.1 and Remark 1.5 · pp.2–3

For purity, only the receiving argument and its stated hypotheses were checked; the original SGA1 statement was not rechecked.

This completes the lower bound involving variation in Theorem 1.1. Corollary 1.2 uses this output with empty boundaries.

WV, Theorem 1.1 / Corollary 1.2 · p. 2; §7.4 · p. 59

6. Corollary 1.2: take empty boundaries

For projective X, Y , use empty compactification boundaries. The geometric generic fiber and its field-of-definition invariant are unchanged.

01 Choose empty boundaries in the projective setting

$$f : X \longrightarrow Y, \quad \kappa(Y) \geq 0, \quad D_X = D_Y = 0.$$

Smooth connected complex projective varieties; surjective; connected fibers.

Use $\bar{\kappa}(X) = \kappa(X)$ and $\bar{\kappa}(Y) = \kappa(Y)$; the field invariant is unchanged.

Thm. 1.1; Cor. 1.2 and proof, p.2

02 Specialize with the same whole-fiber variation

$$\kappa(X) \geq \kappa(F) + \max\{\kappa(Y), \text{Var}(f)\}.$$

Empty boundaries specialize the logarithmic inequality to the projective inequality

[Open figure ↗](#) [TeX source ↓](#)

1. Replace logarithmic by ordinary Kodaira dimensions

With $D_X = D_Y = 0$, one has $\bar{\kappa}(X) = \kappa(X)$ and $\bar{\kappa}(Y) = \kappa(Y)$. The hypothesis $\kappa(Y) \geq 0$ is exactly the base hypothesis of Theorem 1.1, so its inequality gives Corollary 1.2 immediately.

[Proof of Corollary 1.2 · p.2](#), [Section 7.4 · p.59](#)

The ordinary Iitaka–Viehweg inequality applies to projective fibrations over a base of nonnegative Kodaira dimension. The additional relative-Iitaka route is not a premise of this proof.

[WV, Corollary 1.2 · p. 2](#); [§9 introduction · p. 69](#)

7. Corollary 1.3: good models for a smooth family

This route uses smoothness of the family and non-uniruledness of every closed fiber. It does not enter the proof of the main theorem.

01 Use non-uniruledness of every closed fiber

$f : U \rightarrow V$ smooth projective, F_v non-uniruled (v closed).

Smooth connected complex quasi-projective varieties; surjective; connected fibers.

Apply the contrapositive of the projective uniruledness criterion.

[\[BDPP\] Cor. 0.3, p.2](#)

02 Make the canonical divisor pseudo-effective

K_{F_v} pseudo-effective (v closed).

Apply smooth canonical good-model existence to each closed fiber.

[\[LA\] Cor. 11.2, pp.73–74](#) [Cor. 1.3 proof, p.5](#)

03 Obtain a semiample model and an exceptional comparison

$p^*K_{F_v} = q^*K_{M_v} + E$, $E \geq 0$, E q -exceptional.

K_{M_v} semiample.

Compare discrepancies on a common resolution containing any prime exceptional over M_v .

[Cor. 1.3 proof, p.5](#)

04 Check the canonical good model needed by Taji

$a(P, M_v) = a(P, F_v) + \text{coeff}_P E \geq 0$.

Use smoothness and good models of the fibers.

The case $\bar{\kappa}(V) = 0$ also has variation zero. This route is not used in the main proof.

[\[Taj\] Thms. 1.1–1.2 and variation, p.2](#)

05 Bound the same whole-fiber variation

$\bar{\kappa}(V) = -\infty \Rightarrow \text{Var}(f) < \dim V$,

$\bar{\kappa}(V) \geq 0 \Rightarrow \text{Var}(f) \leq \bar{\kappa}(V)$,

V Campana-special $\Rightarrow \text{Var}(f) = 0$.

Non-uniruled closed fibers acquire good models from LA, allowing application of Taji

[Open figure ↗](#) [TeX source ↓](#)

1. Obtain canonical good models from LA's comparison

For every closed fiber F_v , BDPP makes K_{F_v} pseudo-effective. LA Corollary 11.2 supplies a semiample K_{M_v} and a common-resolution identity

$$p^*K_{F_v} = q^*K_{M_v} + E, \quad E \geq 0, \quad E \text{ is } q\text{-exceptional}.$$

For every prime P exceptional over M_v ,

$$a(P, M_v) = a(P, F_v) + \text{coeff}_P E \geq 0.$$

Thus M_v is canonical and gives a good minimal model in the sense used by Taji.

Proof of Corollary 1.3 · p.5 · BDPP, Corollary 0.3 · p.2 · LA, Corollary 11.2 · pp.73–74

2. Apply Taji with whole-fiber variation

For smooth projective families whose fibers have good minimal models, [Taj Theorems 1.1–1.2](#) supply isotriviality over special bases and the two variation bounds according to the sign of the base's logarithmic Kodaira dimension. Taji's definition on p.2 also uses a birational field of definition of the entire fiber. When $\bar{\kappa}(V) = 0$, combine the bound with $\text{Var}(f) \geq 0$.

Corollary 1.3 · p.5 · Taj, Theorems 1.1–1.2 and definition of variation · p.2

For a smooth family with $\kappa(F) \geq 0$ and $\bar{\kappa}(V) \geq 0$, comparing [RA Corollary 1.2's](#) additivity with [Theorem 1.1](#) gives the same upper bound. This alternative does not cover the negative-base branch or all special bases.

Comparison paragraph · p.5 · RA, Corollary 1.2 · p.2

A base of logarithmic Kodaira dimension zero gives birational isotriviality. For the negative-base and special-base conclusions, retain the hypotheses of the LA–Taji route.

WV, Corollary 1.3 · p. 5

8. Additional results in Sections 8–9

The main proof ends in Section 7.4. [Theorem 8.1](#) replaces the Hodge contribution by an arbitrary nef pullback $M = h^*A$, for morphisms $g : W \rightarrow Y$ and $h : W \rightarrow Z$, an effective SNC rational boundary $B \geq (g^*D_Y)_{\text{red}}$, and $\kappa(Y, K_Y + D_Y) \geq 0$. The varieties are smooth complex projective, both morphisms are surjective with connected fibers, D_Y is reduced SNC, the coefficients of B lie in $[0, 1]$, and here $d = \dim(W/Y)$. If $K_Z + aA$ is big for all sufficiently large rational a and $K_W + B + M$ is big over Y , it asserts a second Iitaka base U_2 (called U in the manuscript) satisfying

$$\kappa(W, K_W + B + M) = \dim U_2 = d + \dim T_0, \quad \kappa(Y, K_Y + D_Y) \leq \dim T_0.$$

The final interpolation compares $P_{U_2}(c)$ and $P_{U_2}(a)$ on one fixed model. This reading covered the statement and the final proof passage, not Section 8's intermediate volume and effective finite-group degree estimates.

[Theorem 8.1](#) · pp.59–60, final proof passage · p.69

Section 9 applies this construction to the ordinary relative Iitaka fibration. It cites [OI Proposition 2.7](#), [Lemma 7.1](#), [Corollary 7.2](#), and [Lemma 7.4](#) for section comparison and the reduced-root entire top line. [Corollary 9.4](#) identifies the absolute Iitaka field and the intermediate field, recovering $\mathbb{C}(T_0) = \mathbb{C}(Y) \cap \mathbb{C}(U_2)$ inside $\mathbb{C}(X)$ before applying [Proposition 7.3](#). The additional [OI statements](#) and their use in [Proposition 9.1](#) were compared. Their full proofs and the complete arguments of §§8–9 were not independently verified.

[Proposition 9.1](#), [Corollaries 9.2–9.4](#) · [pp.69–72](#)

9. Which papers supply which steps

Input

[OI Corollary 6.2](#), pp.40–41

Use and role in this manuscript

[Theorem 2.1](#), [Section 2](#): logarithmic lower bounds and zero-dimensional Iitaka systems

Relation and comparison scope

direct; input statement and applications compared

Input

[OI Theorem 3.1](#), pp.17–18

Use and role in this manuscript

[Theorem 3.8](#), [Propositions 3.9](#) and [3.12](#): restricted adjoint comparison

Relation and comparison scope

direct; integral ambient and complex-summand hypotheses compared

Input

[Fuj Theorem 1.1](#), pp.1–2

Use and role in this manuscript

[Lemma 3.11](#), pp.22–23: sections after a big base twist

Relation and comparison scope

direct; lc, Cartier multiple, and projective-base hypotheses compared

Input

[Del Corollary 4.2.8\(iii\)\(b\)](#), printed p.47

Use and role in this manuscript

Corollaries 3.14–3.16: finite character of a rank-one subsystem

Relation and comparison scope

direct; statement and use compared; invariant-cycle citation separately unverified

Input

[DHP Theorem 4.1](#), pp.20–21

Use and role in this manuscript

Lemma 4.4, pp.30–31: extension under a putatively nonzero restriction to the zero divisor

Relation and comparison scope

direct; conditions (19)–(23) compared with the application

Input

[PT Theorem 4.2.7](#) and [Remark 4.3.1](#), pp.36–37

Use and role in this manuscript

Lemma 4.6, pp.34–35: relative Bergman weight and uniform upper bound

Relation and comparison scope

direct; statement and use compared

Input

[BCHM Theorem 1.2\(2\)](#), p.5

Use and role in this manuscript

Section 4.6, p.40, Section 6, p.48: big klt adjoint models

Relation and comparison scope

direct; model-existence statement and uses compared; auxiliary ample-model locators unverified

Input

KP Theorem 9.9, p.50

Use and role in this manuscript

Proposition 5.3, p.46 → Proposition 6.2: lower bound for marked-model variation

Relation and comparison scope

direct; auxiliary base divisor $M = 0$ and klt big generic pair compared

Input

Han Theorems 2.1–2.2

Use and role in this manuscript

Lemma 7.1, p.53: birational identity component and cocycle

Relation and comparison scope

direct; receiving citation and argument read; original statements not compared

Input

SGA1 X, Théorème 3.1

Use and role in this manuscript

Proposition 7.3, p.57: purity for a finite normalization unramified in height one

Relation and comparison scope

direct; hypotheses as listed by the receiving paper checked

Input

Lan Theorem 1.1 and Remark 1.5, pp.2–3

Use and role in this manuscript

Proposition 7.3, pp.57–58: descent of finite étale covers and morphisms

Relation and comparison scope

direct; characteristic-zero specialization and use compared

Input

BDPP Corollary 0.3, LA Corollary 11.2, Taj Theorems 1.1–1.2

Use and role in this manuscript

Corollary 1.3, p.5

Relation and comparison scope

consequence; input statements and interfaces compared; not used in the main theorem

Input

PH Lemma 8.4 and Theorem 8.1

Use and role in this manuscript

Remark 3.17, p.25

Relation and comparison scope

alternative; input statements and use compared, including checks recorded in the PH article;
input proofs unverified

Input

OI Proposition 2.7, Lemma 7.1, Corollary 7.2, Lemma 7.4

Use and role in this manuscript

Proposition 9.1, pp.69–70

Relation and comparison scope

consequence (directly used by the additional result); statements and uses compared, full proofs
unverified

Unrecorded relations remain unexamined. Comparing an input's statement with its application is distinct from verifying the input theorem's entire proof.

10. Return to the sources

- Embedded parameter field: Proposition 2.10, equation (2.12), pp.12–14.
- Actual minimum versus all-valuation threshold: Propositions 3.3 and 3.5, pp.17–19, returning in Lemma 6.4's multiplicity-one minimizer.

- First constancy statement: Proposition 3.12, Theorem 4.1, Lemmas 4.6–4.7, Proposition 4.8, pp.23–43.
- Second constancy statement and the whole field: Proposition 6.2, Lemma 6.4, Propositions 7.2–7.3, pp.47–59.

This reading covered the main statements and corollaries, Section 2's parameter field, Section 3's root cover, orders, and restricted triviality, the principal analytic constancy argument of Section 4, and the marked models and descent in Sections 5–7. The external interfaces compared are listed in the table. This is an account of the argument's route, not an independent verification of every inference.

Remaining work includes the full regularization and L^2 estimates of Section 4, external Hodge-extension and boundary-growth inputs, Hanamura's original statements, the original SGA1 purity statement, Section 8's intermediate proofs, and full proofs of Section 9's additional [OI](#) inputs. The full proofs of external inputs, expert review, and formal verification have not been completed. The separate scope for Sections 8–9 is stated above.

The catalogue-033 manuscripts, including [WV](#) and [OI](#), use commit

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. [LA](#) uses `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, matching the existing catalogue-034 article. Page locators refer to the respective PDFs; Deligne's printed pagination is additionally indicated.

For the additional route, the statements of [OI](#) Proposition 2.7, Lemma 7.1, Corollary 7.2 and Lemma 7.4 were compared with their use in Proposition 9.1. Complete valuation comparisons, all BFMT interfaces and the full proofs of §§8–9 remain unverified.

[OI](#), Proposition 2.7 · Lemma 7.1 · Corollary 7.2 · Lemma 7.4 · pp.10–17, 43–47 ·
[WV](#), Proposition 9.1 · pp.69–70

The reverse logarithmic Kodaira inequality and additivity

Sections retaining a prescribed zero, and removal of the period twist

For projective pairs whose total space and all boundary strata are smooth over the open base, the manuscript proves an upper bound for logarithmic Kodaira dimension. Its central construction, Proposition 1.3, turns a total-space pluriform into a pluriform on the original base and transfers vanishing on a specified fiber to vanishing at the specified base point. We first follow the independent upper-bound argument and then treat additivity, which additionally uses the lower bound from OI.

Manuscript author: OpenAI · 2026-09-26 · 51 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

Theorem 1.1 — Reverse logarithmic Kodaira inequality

Let X, Y be smooth connected projective complex varieties and let $f : X \rightarrow Y$ be surjective with connected fibers. Let $E \subset X$ and $D \subset Y$ be reduced SNC divisors, either of which may be zero, satisfying

$$\mathrm{Supp}(f^*D) \subseteq \mathrm{Supp}(E).$$

Set $V = Y \setminus \mathrm{Supp} D$. Assume that X , and every irreducible component of every nonempty intersection of components of E , restricted to $f^{-1}(V)$, are smooth over V . For very general $y \in V$, put $F = X_y$, $E_F = E|_F$, and write

$$L_X = K_X + E, \quad L_Y = K_Y + D, \quad L_F = K_F + E_F.$$

Then

$$\kappa(X, L_X) \leq \kappa(Y, L_Y) + \kappa(F, L_F).$$

If either term on the right is $-\infty$, then $H^0(X, mL_X) = 0$ for every integer $m > 0$. A point has Kodaira dimension 0, and $(-\infty) + a = -\infty$, including $a = -\infty$. The boundary condition concerns supports and does not discard the multiplicities of f^*D . No smoothness over D , abundance, semiample, or good minimal model is assumed.

Theorem 1.1 and (1.1)–(1.2) · p. 2

Corollary 1.2 — Logarithmic additivity

For the same pairs, morphism, and stratum-smoothness assumptions as in Theorem 1.1, with the same $-\infty$ convention,

$$\kappa(X, K_X + E) = \kappa(Y, K_Y + D) + \kappa(F, K_F + E_F).$$

This combines the upper bound with [OI, Corollary 6.2]. The proof of the upper bound in Theorem 1.1 does not use OI.

Corollary 1.2 · p. 2; §7.6 · p. 50 · OI, Corollary 6.2 · pp. 40–41

Proposition 1.3 — A section on the original base with a prescribed zero

Suppose f, E, D satisfy the hypotheses of Theorem 1.1 and $0 \neq s \in H^0(X, mL_X)$ for some integer $m > 0$. Then $\kappa(Y, L_Y) \geq 0$. If, moreover, $\kappa(Y, L_Y) = 0$ and $s|_{X_y} = 0$ for some $y \in V$, there are an integer $a > 0$ and a section

$$0 \neq \tau \in H^0(Y, aL_Y), \quad \tau(y) = 0.$$

The point y may be any such point of the original V : it need not lie in the smooth locus of the cyclic-cover family constructed from s .

Proposition 1.3 · p. 3

References on the arrows

Unprefixed numbers refer to RA. The diagrams retain the Proposition / Lemma / Corollary labels at the statements. Some later cross-references in the manuscript instead call these results “Theorem.” GitHub preview links do not guarantee navigation to a PDF page; page numbers appear explicitly in the labels.

- OI: Orbifold and logarithmic Iitaka subadditivity, pinned 2026-09-26 manuscript.
- FF14: Fujino–Fujisawa, Variations of mixed Hodge structure and semipositivity theorems, author manuscript dated 2014-03-17. FF17 is their supplement dated 2017-07-07.
- BBT: Bakker–Brunenbarbe–Tsimmerman, o-minimal GAGA and a conjecture of Griffiths, arXiv v3.

- **FG:** Fujino–Gongyo, On the moduli b-divisors of lc-trivial fibrations, published version, 2014.
- **CP:** Campana–Păun, Foliations with positive slopes and birational stability of orbifold cotangent bundles, checked in arXiv v4.
- **BDPP:** Boucksom–Demailly–Păun–Peternell, The pseudo-effective cone of a compact Kähler manifold and varieties of negative Kodaira dimension, arXiv v1.

2. Proof of Theorem 1.1 — Reduction by Kodaira dimension

Proof route. First derive the upper bound with Proposition 1.3 as input. Its proof is then expanded into constructing sections from a Hodge vector and removing the twist for a positive-dimensional period quotient. OI enters only in the final additivity corollary.

Write $\kappa_X, \kappa_Y, \kappa_F$ for the logarithmic Kodaira dimensions of the respective pairs. Given Proposition 1.3, the upper bound follows from restriction of sections and the logarithmic Iitaka fibration of the base.

Theorem 1.1 Compare sections in each Kodaira-dimension case

$$\kappa_X = \kappa(X, L_X), \quad \kappa_Y = \kappa(Y, L_Y), \quad \kappa_F = \kappa(F, L_F).$$

A Fiber of negative-infinite dimension

$$\kappa_F = -\infty.$$

A nonzero total section restricts nontrivially to a very general fiber.

§2.2, p. 5

Restrict to a very general fiber

$$H^0(X, mL_X) = 0 \quad (m > 0).$$

B Base of negative-infinite dimension

$$\kappa_Y = -\infty.$$

A nonzero total section forces $\kappa_Y \geq 0$.

Prop. 1.3, p. 3 §2.2, p. 5

Contradict base nonvanishing

$$H^0(X, mL_X) = 0 \quad (m > 0).$$

C Base of Kodaira dimension zero

$$\kappa_Y = 0, \quad h^0(Y, aL_Y) \leq 1.$$

Choose $y \in V$ outside all base zero divisors.

If $s|_{X_y} = 0$, a base section vanishes at the original y , contradicting its choice.

Prop. 1.3, p. 3

§2.3, (2.3)–(2.5), pp. 5–6

Injective restriction in every degree

$$H^0(X, mL_X) \hookrightarrow H^0(F, mL_F).$$

$$\kappa_X \leq \kappa_F.$$

D Base of positive Kodaira dimension

$$k = \kappa_Y > 0, \quad q : Y' \rightarrow T, \quad \dim T = k.$$

$$\kappa(G, K_G + D'|_G) = 0.$$

The induced pairs retain stratum smoothness. Bound the image jointly with $q \circ f'$.

§2.4, pp. 6–7

Apply the zero-base case on litaka fibers

$$H = (q \circ f')^{-1}(t), \quad H \rightarrow G.$$

$$\begin{aligned} \kappa_X &\leq k + \kappa(H, K_H + E'|_H) \\ &\leq k + \kappa_F. \end{aligned}$$

END The same upper bound in all cases

$$\kappa_X \leq \kappa_Y + \kappa_F, \quad (-\infty) + a = -\infty.$$

Upper bound: the two negative-infinity cases, restriction injectivity over a zero-dimensional litaka image, and the positive-base reduction

Open figure ↗ TeX source ↓

1. Treat both negative-infinity cases as vanishing statements

If $\kappa_F = -\infty$, the nonvanishing locus of a hypothetical nonzero total-space section has image containing a dense open subset of the base. Restriction and adjunction would give

$H^0(F, mL_F) \neq 0$ on a very general fiber, a contradiction. If $\kappa_Y = -\infty$, Proposition 1.3 instead contradicts the existence of a total-space section by producing base nonvanishing. Both arguments exclude sections in every positive degree.

§2.2 · p. 5

2. Obtain restriction injectivity in every degree at one fiber

When $\kappa_Y = 0$, one has $h^0(Y, aL_Y) \leq 1$. Each degree therefore contributes at most one zero divisor of a nonzero section. Choose y outside their countable union, also making all fiber plurigenera attain their very general values. A nonzero s in the restriction kernel would, by Proposition 1.3, produce a nonzero base section vanishing at y , contradicting this choice. Hence

$$H^0(X, mL_X) \hookrightarrow H^0(F, mL_F) \quad (m > 0), \quad \kappa_X \leq \kappa_F.$$

The essential point is that Proposition 1.3 applies at the specified point of the **original** V , independently of the auxiliary open set attached to each section. Finite generation of the section rings is unnecessary.

§2.3 and (2.3)–(2.5) · pp. 5–6

3. Reduce positive base dimension to zero on an Iitaka fiber

For $k = \kappa_Y > 0$, resolve the logarithmic Iitaka map to $q : Y' \rightarrow T$ with $\dim T = k$. Use strict-transform boundaries plus reduced exceptional divisors and make a compatible resolution $f' : X' \rightarrow Y'$; logarithmic section spaces are unchanged. A very general fiber G of q has $\kappa(G, K_G + D'|_G) = 0$. The induced morphism from $H = (q \circ f')^{-1}(t)$ to G retains the required smoothness with its boundary strata, so the preceding case applies:

$$\kappa(H, K_H + E'|_H) \leq \kappa_F, \quad \kappa_X \leq \dim T + \kappa(H, K_H + E'|_H) \leq k + \kappa_F.$$

The last upper bound is the image-dimension estimate for a total-space linear system combined with $q \circ f'$. It includes the case where G is a point. If H has Kodaira dimension $-\infty$, restriction excludes all total-space sections instead.

§2.1 and (2.1) · p. 5; §2.4 · pp. 6–7

Section 5 combines this independent upper bound with OI's logarithmic lower bound. Its key input is Proposition 1.3, developed next.

RA, §2 · §7.6 · pp. 5–7, 50

3. Proof of Proposition 1.3 — From a Hodge vector to a base section

Fix a nonzero section $\mathfrak{s}$. The constructed Hodge vector need not generate the whole highest Hodge bundle. The argument therefore retains its coefficients and the compact flags determined by its projective direction. Below, $\mathcal{E}_0$ denotes the highest Hodge bundle, distinct from the boundary E .

START Fix one total-space pluriform

$$0 \neq s \in H^0(X, mL_X), \quad V = Y \setminus \text{Supp } D.$$

Take the root class and a nonzero pure weight quotient. Semistable comparison and weight subbundles extend each coefficient regularly.

Prop. 3.1, Lem. 3.2, pp. 8–13 [FF14] §4.9, pp. 24–25

[FF17] pp. 1–2

01 Keep the original coefficient lattice in a pure weight quotient

$$u : \mathcal{O}_Y(-L_Y) \dashrightarrow \mathcal{E}^d, \quad \theta^j(u) : \mathcal{O}_Y(-L_Y) \dashrightarrow \mathcal{E}^{d-j} \otimes \Omega_Y^1(\log D)^{\otimes j}.$$

Use the original source frame after blowing up. Divide out the root's scalar factor before weight projection; homogeneous operations preserve the zero.

Lem. 3.3, (3.15), pp. 13–14

02 Preserve the specified zero in the original frame

$$u_* : \mathcal{O}_Y(-aL_Y) \dashrightarrow \mathcal{E}_0, \quad y \in V.$$

$$s|_{X_y} = 0 \implies \text{ord}_\tau(u_*) \geq ae/m > 0 \quad (t = \tau^e, m \mid e).$$

Algebraize the integral Lie-period image and add the compact marks.

Prop. 4.1, pp. 14–15 Lem. 4.7, pp. 19–20

[BBT] Thm. 1.1, p. 1

03 Retain the full Lie period and compact flags

$$Y \dashrightarrow S, \quad \mathcal{E}_0 \text{ comes from a variation on } S.$$

Constant-frame coefficients are regular.
Positive exceptional order retains the zero.

§7.1, p. 43

Use the exact section comparison and the numerical removal in the next diagram.

Prop. 6.3, pp. 40–43

Prop. 7.7, p. 49

A S a point: select a nonzero coordinate

$$0 \neq \tau \in H^0(Y, aL_Y).$$

$$s|_{X_y} = 0 \implies \tau(y) = 0.$$

B $\dim S > 0$: remove the twist

$$L_W \text{ big} \implies$$

$$\kappa(Y, L_Y) \geq \dim W \geq \dim S > 0.$$

END Base nonvanishing and a prescribed zero when the base has Kodaira dimension zero

$$\kappa(Y, L_Y) \geq 0; \quad \kappa(Y, L_Y) = 0 \implies S = \text{pt.}$$

Prop. 7.1, p. 43 Prop. 1.3: proof, p. 50

Section construction: preserve the original coefficient lattice and zero test, then separate point and positive-dimensional period quotients

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1. Keep new cyclic-cover discriminants out of the base pole divisor

Taking an m th root of s produces a logarithmic mixed Hodge class; projection to a suitable pure weight quotient remains nonzero. The coefficients of the resulting $u : \mathcal{O}_Y(-L_Y) \dashrightarrow \mathcal{E}^d$ and its Higgs iterates are measured in the original $\Omega_Y^1(\log D)$ frames. Proposition 3.1 proves their regularity in the unipotent Hodge extension on each prescribed higher model.

In transverse-curve tests, the cotangent tensor frames remain frames of an external pullback bundle. They are not replaced by differentials on the testing curve and then divided back. Thus an extra cyclic-cover discriminant inside the original V introduces no new allowed poles. Regular projection to a pure weight quotient uses the subbundle property of Lemma 3.2, with the semistable logarithmic de Rham/canonical-extension comparison from FF14 and the strictness supplement FF17.

Proposition 3.1 and Lemma 3.2 · pp. 8–13 · FF14, §4.9 and Lemma 4.10, Step 1 · pp. 24–31 · FF17 · pp. 1–2

2. Retain the original source frame when blowing up the specified point

If $s|_{X_y} = 0$, its coefficient near the smooth relative SNC fiber belongs to $\mathfrak{m}_y \mathcal{O}_X$. Blow up y , take a transverse parameter t at the exceptional divisor, and set $t = \tau^e$ with $m \mid e$. In the original L_X frame, the root r satisfies

$$(r/\tau^{e/m})^m = s/\tau^e.$$

The right-hand side is regular, and normality makes $r/\tau^{e/m}$ regular. The weight-subbundle property in Lemma 3.2 allows this scalar factor to be divided out before projection. After a homogeneous tensor operation of degree a , the vanishing order is still at least $ae/m > 0$ in the pulled-back original $\mathcal{O}_Y(-aL_Y)$ frame. Replacing it by the new logarithmic canonical frame on the blowup would lose the prescribed-point information.

Lemma 3.3 and (3.15) · pp. 13–14

3. Extract a scalar coordinate for a point quotient; prove positivity otherwise

Proposition 4.1 constructs a homogeneous replacement $u_* : \mathcal{O}_Y(-aL_Y) \dashrightarrow \mathcal{E}_0$ and a connected quotient $\mathcal{S}$ retaining the full Lie period and compact flags. [BBT, Theorem 1.1] algebraizes the period image of the full Lie variation, in which an integral lattice has been retained. The quotient is not obtained merely from the determinant of the highest Hodge bundle.

If $\mathcal{S}$ is a point, the descended variation is constant. In a constant frame, each coordinate of u_* is a rational section of aL_Y . Divisorial regularity and normality make it global; choose a nonzero coordinate as τ . The positive exceptional order from the preceding step is the vanishing order at

the original y , so $s|_{X_y} = 0$ implies $\tau(y) = 0$. When $\dim S > 0$, the next section gives $\kappa_Y > 0$. Consequently only the point case can occur for $\kappa_Y = 0$, proving both conclusions of Proposition 1.3.

Proposition 4.1 · pp. 14–15; Lemma 4.7 · pp. 19–20; §7.1 and Proposition 7.1 · p. 43; §7.5 · pp. 49–50 ·
BBT, Theorem 1.1 · p. 1

A base section vanishing at the specified point makes restriction to one fiber injective in every degree when the base has Kodaira dimension zero. The next section handles the positive-dimensional period quotient.

RA, §2.3 · proof of Proposition 1.3 · pp. 5–6, 50

4. A positive-dimensional period quotient — Remove the twist and recover sections

Assume $\dim S > 0$. Write $Y \xrightarrow{x} W \xrightarrow{h} S$ on suitably prepared models; the section comparison always returns to a fixed original pair. The notation $\geq_{\text{pe}}$ means that the difference is pseudoeffective.

START Start from a positive-dimensional marked quotient

$$\dim S > 0, \quad Y \xrightarrow{x} W \xrightarrow{h} S, \quad u_* = \omega x^* u_W.$$

Factor by the generic-fiber generator and extract original source valuations.
The minimum detects all poles and preserves section ratios.

[Lem. 6.1–6.2, pp. 37–40](#) [Prop. 6.3, pp. 40–41](#)

01 Identify complete section spaces with the original base

$$L_W = K_W + \sum_A t_A A, \quad t_A = \min_{I \mapsto A} \frac{a^{-1} \operatorname{ord}_I \omega + d_I}{m_I}.$$

$$H^0(W, \ell L_W) \simeq H^0(Y_0, \ell L_{Y_0}) \quad (\ell \text{ sufficiently divisible}), \quad L_W \text{ big over } S.$$

Allow negative coefficients in Δ and verify rank one for the discrepancy sheaf.

[Prop. 6.3, \(6.12\)–\(6.13\), pp. 41–42](#)

[\[FG\] Def. 3.2, Thm. 3.6, PDF pp. 5, 7](#)

02 Obtain a nef moduli part from an auxiliary lc-trivial subpair

$$L_W = K_W + B_W + M_W, \quad 0 \leq B_W \leq 1, \quad M_W \text{ nef.}$$

Remove D_0 by boundary rank loss; compare curvature-null directions of the tails.

[Prop. 5.3, 5.5, pp. 30–34](#) [Prop. 5.6, pp. 34–36](#)

03 Compare positivity from periods and Higgs tails

$$K_S + bH \text{ big}, \quad P = -\sum_i (i+1) \det G_i \text{ nef}, \quad \tilde{H} = \pi^* h^* H.$$

$$\nu(P + \tilde{H}) = \nu(P).$$

Transfer fixed Higgs maps to orbifold cotangent tensors. Movable-curve estimates
force threshold zero; perturbation also gives relative pseudoeffectivity.

[Lem. 7.4–7.6, pp. 46–49](#)

[\[CP\] Thm. 1.3, 3.4, pp. 3, 12–14](#) [\[BDPP\] Thm. 2.2, p. 6](#)

04 Obtain a comparison uniform under boundary perturbation

$$Q\pi^* L_W \geq_{\text{pe}} P \quad (Q > 0), \quad L_W - h^* K_S \geq_{\text{pe}} 0.$$

First $L_W + bh^* H$ is big. Nef subtraction removes the twist; the complete
section-space comparison returns sections to the original base.

[Prop. 7.2, Lem. 7.3, pp. 44–45](#) [Prop. 7.7, p. 49](#)

05 Subtract the period twist

$$(1 + tQ)\pi^* L_W \geq_{\text{pe}} A' + tP - b\tilde{H} \text{ big} \quad (t \gg 0).$$

$$L_W \text{ big} \implies \kappa(Y, L_Y) \geq \dim W \geq \dim S > 0.$$

Positive-dimensional quotient: the exact section lattice, nef moduli and Higgs-tail lines, and numerical removal of the period twist

Open figure ↗ [TeX source](#) ↓

1. Construct a divisor representing the complete section spaces

The coefficient system is constant on the generic fiber over S , so a nonzero logarithmic section exists there and its Iitaka dimension is nonnegative. Form its relative Iitaka fibration. The unique degree- a generator ω on the generic x -fiber J factors $u_* = \omega x^* u_W$. For a prime A on W , define

$$t_A = \min_{I \rightarrow A} \frac{a^{-1} \operatorname{ord}_I(\omega) + d_I}{m_I}, \quad m_I = \operatorname{ord}_I(x^* A), \quad d_I = \operatorname{coeff}_I D_Y, \quad L_W = K_W + \sum_A t_A A.$$

Here D_Y is the reduced boundary on this model. If a prime on the original source is contracted to codimension at least two on a temporary base, its restricted valuation is first extracted on the base. The minimum rule then yields, in all sufficiently divisible degrees ℓ , an isomorphism

$$H^0(W, \ell L_W) \xrightarrow{\sigma \mapsto \omega^{\ell/a} x^* \sigma} H^0(Y_0, \ell L_{Y_0})$$

preserving section ratios. The fixed model Y_0 has the same section spaces as the original (Y, D) . Moreover, L_W is big over S .

Lemmas 6.1–6.2 and (6.5)–(6.6); Proposition 6.3 and (6.8) · pp. 37–41

2. Check the auxiliary subpair's rank condition before applying b-nefness

The manuscript sets $\Delta = -a^{-1} \operatorname{div}_{aK_{Y/W}}(\omega) + x^* \sum t_A A$, obtaining $K_Y + \Delta \sim_{\mathbb{Q}} x^* L_W$; negative coefficients of Δ are allowed. The logarithmic zero divisor Z_J on the generic fiber has Iitaka dimension zero. Sections of the discrepancy sheaf are bounded above by sections of $\mathcal{O}_J(NZ_J)$ and below by the constant section, giving

$$\operatorname{rank} x_* \mathcal{O}_Y([\mathbf{A}^*(Y, \Delta)]) = 1.$$

[FG, Definition 3.2 / Theorem 3.6] now applies. On a suitable model,

$$L_W = K_W + B_W + M_W, \quad 0 \leq B_W \leq 1, \quad M_W \text{ nef.}$$

The lower bound on B_W comes from testing the discriminant threshold at the same prime attaining the minimum for t_A . The external input supplies b-nefness, without requiring semiamplessness of the moduli part.

Proposition 6.3 and (6.9)–(6.15) · pp. 41–43 ·

FG, Definition 3.2 and Theorem 3.6 · PDF pp. 5, 7 (printed pp. 1724, 1726)

3. Account for the period directions in the Higgs-tail numerical dimension

On each component of the boundary D_0 with nonidentity Lie monodromy, the period line satisfies $\nu(H|_{D_i}) < \nu(H)$. Comparing growth orders of sections removes this boundary and makes $K_S + bH$ big. Let G_i be generated by all contractions of the i th Higgs iterate of u_W . On a unipotent cover followed by resolution, $\pi : \widehat{W} \rightarrow W$, put

$$P = - \sum_i (i+1) \det G_i, \quad \tilde{H} = \pi^* h^* H.$$

Proposition 5.6 proves that P is nef and $\nu(P + \tilde{H}) = \nu(P)$. A curvature-null direction fixes the chosen line and its compact flags; an orbit calculation and the representation-theoretic Lemma 5.7 then show that it fixes the whole highest Hodge space. The chosen vector is not assumed to span that space.

Proposition 5.3 · pp. 30–33; Lemma 5.4 and Proposition 5.5 · pp. 33–34; Proposition 5.6 and (5.11)–(5.18) · pp. 34–36

4. Obtain a fixed comparison from estimates uniform in the boundary

Lemma 7.4 uses the minimizing prime for t_A and the discriminant estimate together to transfer Higgs determinants into orbifold cotangent tensors. Represent $M_W + \epsilon A$ by a general effective Λ_ϵ and apply [CP, Theorem 1.3] whenever $K_W + B_W + \Lambda_\epsilon \sim_{\mathbb{Q}} L_W + \epsilon A$ is pseudoeffective. [BDPP, Theorem 2.2] converts movable-curve degree inequalities into

$$C(L_W + \epsilon A) + RL_W \geq_{\text{pe}} P_W,$$

with $C \geq 0$ and $R > 0$ independent of ϵ . A positive pseudoeffective threshold t_0 would satisfy $t_0 \leq Ct_0/(C+R) < t_0$, so $t_0 = 0$. Taking limits and using $\pi^* P_W \geq_{\text{pe}} P$ gives $Q\pi^* L_W \geq_{\text{pe}} P$ for the fixed $Q = C + R > 0$. Applying [CP, Theorem 3.4] to the same perturbed pairs and pushing down from good models also gives $L_W - h^* K_S \geq_{\text{pe}} 0$.

Lemmas 7.4–7.6 and (7.5)–(7.11) · pp. 46–49 · CP, Theorems 1.3, 3.4 and Remarks 3.3, 3.6 · pp. 3, 12–14 · BDPP, Theorem 2.2 · p. 6

5. Subtract the numerical twist and return to the original base

Relative bigness, $L_W - h^* K_S \geq_{\text{pe}} 0$, and bigness of $K_S + bH$ first make $L_W + bh^* H$ big. After pullback, choose an ample lower bound A' and add the fixed comparison from the preceding step:

$$(1 + tQ)\pi^* L_W \geq_{\text{pe}} A' + tP - b\tilde{H}.$$

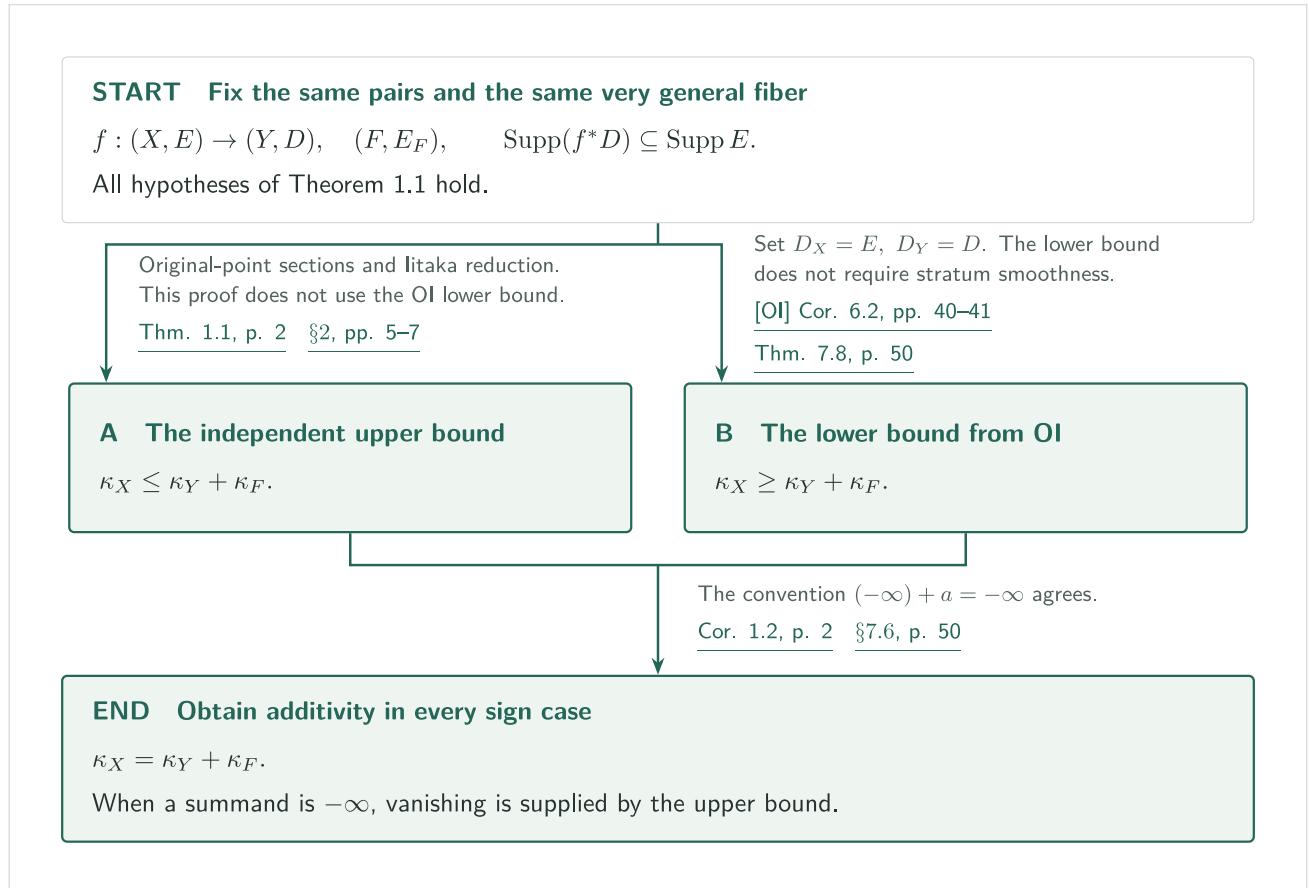
In Lemma 7.3, the equality $\nu(P + \tilde{H}) = \nu(P)$ makes the loss term in the nef Morse inequality have smaller degree in t than the supply term. The right-hand side is therefore big for $t \gg 0$, and Proposition 7.2 makes L_W itself big. The **complete section-space isomorphism** from Step 1 returns these systems to the original base and gives $\kappa_Y \geq \dim W \geq \dim S > 0$. No equality between numerical and Iitaka dimensions of an arbitrary pseudoeffective divisor is assumed.

Proposition 7.2 and Lemma 7.3 · pp. 44–45; Proposition 7.7 · p. 49; proof of Proposition 1.3 · p. 50

Positive base Kodaira dimension excludes this branch when the base has Kodaira dimension zero. This completes Proposition 1.3 and returns to the upper-bound argument in Section 2.

5. Proof of Corollary 1.2 — Add the lower bound

Only here does the other catalogue-033 manuscript [OI](#) enter. The notation $\kappa_X, \kappa_Y, \kappa_F$ and the $-\infty$ convention are unchanged from Section 2.



Additivity: combine the independent upper bound with OI logarithmic subadditivity for the same pairs and the same fiber

[Open figure ↗](#) [TeX source ↓](#)

1. Match the boundary condition and the very general fiber

Apply [OI, Corollary 6.2] with $D_X = E$ and $D_Y = D$. Smooth connected projective complex varieties, connected fibers, reduced SNC boundaries, and support inclusion are all supplied by Theorem 1.1. The lower bound does not require stratum smoothness, and its very general pair is exactly (F, E_F) . The manuscript restates it as Theorem 7.8.

Theorem 7.8 and §7.6 · p. 50 · OI, Corollary 6.2 · pp. 40–41

2. Keep the vanishing supplied by the upper bound in the negative-infinity case

When both summands are finite, the two inequalities give equality. If either is $-\infty$, the lower bound is automatic; the content of equality is the vanishing $H^0(X, mL_X) = 0$ supplied by Theorem 1.1. This vanishing is not extracted from the proof in OI.

Theorem 1.1 · p. 2; §2.2 · p. 5; §7.6 · p. 50 · OI, proof of Corollary 6.2 · p. 41

For smooth families with nonnegative fiber and base Kodaira dimensions, comparison with WV Theorem 1.1 gives an alternative upper bound on variation.

WV, comparison after Corollary 1.3 · p. 5

6. Which papers supply which steps

Input

OI, Corollary 6.2 · pp. 40–41

Use in this manuscript

Corollary 1.2 and Theorem 7.8 · pp. 2, 50

Role and checking scope

Input to a consequence: lower bound for the same boundaries and fiber. Compared the statement, short corollary proof, and application; the complete OI proof is not verified.

Input

FF14, §4.9 and Lemma 4.10, Step 1 · pp. 24–31; FF17 · pp. 1–2

Use in this manuscript

Lemma 3.2 · pp. 12–13

Role and checking scope

Hodge extension and regular weight projection. Compared the statements and the local-freeness/strictness interface; the full proofs back to Steenbrink and Deligne are not traced.

Input

BBT, Theorem 1.1 · p. 1

Use in this manuscript

Lemma 4.7 · pp. 19–20

Role and checking scope

Algebraizes the full integral Lie-period image. Compared the input statement and object to which it is applied; compact-mark construction and finite-monodromy descent are not fully independently checked.

Input

FG, Definition 3.2 and Theorem 3.6 · PDF pp. 5, 7

Use in this manuscript

Proposition 6.3 · pp. 41–43

Role and checking scope

b-nefness after checking rank one for the auxiliary subpair, allowing negative coefficients. Compared the definition and theorem with (6.12)–(6.13); input proof not verified.

Input

CP, Theorems 1.3, 3.4 · pp. 3, 12–14

Use in this manuscript

Lemmas 7.5–7.6 · pp. 47–49

Role and checking scope

Orbifold cotangent quotient degrees and relative pseudoeffectivity. Compared perturbed pairs, good models, and pushdown at the application; complete input proofs not verified.

Input

BDPP, Theorem 2.2 · p. 6

Use in this manuscript

Lemma 7.5 and (7.10) · p. 48

Role and checking scope

Passes from movable-curve degrees to pseudoeffectivity. Compared the projective statement and its application; complete input proof not verified.

AS, Theorem A · pp. 1–2 is cited for the connection-form argument in §5.1 and Lemma 5.1 · pp. 26–28. The manuscript supplies the needed semisimple argument itself; we distinguish this as a relation of proof method. Theorem A and the manuscript's stated use were compared, without tracing the full proof in AS. Other relations within catalogue 033 remain unexamined or unrecorded; a missing arrow does not mean absence of dependence.

7. Return to the sources

- §2 · pp. 5–7: all sign cases of the upper bound, using Proposition 1.3 alone.
- Proposition 3.1 and Lemmas 3.2–3.3 · pp. 8–14: the original coefficient lattice and prescribed zero.
- Proposition 4.1 and Lemma 4.7 · pp. 14–15, 19–20; Propositions 5.3, 5.5–5.6 · pp. 30–36: the marked quotient, boundary rank loss, and Higgs tails.
- §6 · pp. 37–43: relative Itaka, extraction of source valuations, exact section comparison, and lc-trivial conditions.
- §7 · pp. 43–50: constant coordinates, twist removal, return to the original base, and addition of the lower bound.

Checked in this draft: the main hypotheses and conclusions and the selected proof passages above, concentrating on §§2–3, the valuation and rank conditions in §6, and the threshold and numerical removal in §7. In §§4–5, the quotient/descent passages, boundary-drop and Higgs-tail statements, their later uses, and selected proof paragraphs were read. External statements and applications were compared as specified in the table.

Not checked: an independent verification of the full homogeneous-replacement/compact-flag construction in §4.1–4.2, all metric estimates in §4.4, or all boundary-degeneration and representation-theoretic calculations in §5. Complete external proofs, recursive verification of

their references, an exhaustive dependency audit, expert review, and formal verification have not been performed. The diagrams organize the manuscript's argument and do not certify its full correctness.

The manuscript and [OI](#) are pinned to commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, version 2026-09-26. Page numbers refer to the linked PDFs; [FG](#) includes a cover, so PDF and printed page numbers are distinguished.

033 / 04

[Read on the website ↗](#)

CATALOGUE 033 · PAPER 04 / 5 Overview

Projective Hodge lines and ordinary Iitaka subadditivity

From rank loss of the highest Hodge line to ordinary Iitaka subadditivity

The manuscript claims ordinary Iitaka subadditivity by combining adjoint positivity for an entire rank-one highest Hodge piece with a canonical bundle formula and weak positivity. Its central mechanism separates the dimension of the full period image from the rank of the highest-line map, removes boundary and finite-quotient ramification, and constructs sections without assuming semiampleness.

Manuscript author: OpenAI · 2026-09-27 · 40 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

The following statements report the manuscript's claims in their original order and with their hypotheses.

Theorem 1.1 — Ordinary Iitaka subadditivity

Let k be an algebraically closed field of characteristic 0, and let $f : X \rightarrow Z$ be a surjective projective morphism with connected fibres between smooth connected projective k -varieties. For its geometric generic fibre F ,

$$\kappa(X) \geq \kappa(F) + \kappa(Z).$$

Use $(-\infty) + a = -\infty$ and $\kappa(\text{point}) = 0$.

[Theorem 1.1 · p. 2](#)

Theorem 1.2 — Projective adjoint positivity

Let Y be a smooth connected projective complex variety, $U \subset Y$ a dense Zariski open subset, and $\mathbb{V}$ a rationally polarized integral pure variation of Hodge structure on U .

Assume that its **entire highest nonzero Hodge filtration step is a line**: $F^{p+1} = 0$ and $\text{rank } F^p = 1$. Let $M \in \text{Pic}(Y) \otimes \mathbb{Q}$. Assume that, after a smooth projective birational modification if needed, a smooth projective alteration $\tau : \hat{Y} \rightarrow Y$ and $c \in \mathbb{Q}_{>0}$ exist such that the pulled-back variation has unipotent monodromy along an SNC boundary and its extended highest Hodge line satisfies

$$\tau^* M \sim_{\mathbb{Q}} c J_{\hat{Y}}.$$

After further smooth projective birational modification, there are a surjective morphism $p : Y \rightarrow S$ with connected fibres onto a smooth projective variety and a nef rational line L such that

$$M \sim_{\mathbb{Q}} p^* L, \quad K_S + jL \text{ is big for every sufficiently large integer } j.$$

If S is a point, the conclusion is $M \sim_{\mathbb{Q}} 0$.

Theorem 1.2 · p. 2

References on the arrows

Unprefixed result numbers refer to [PH](#). The main external inputs use the following keys.

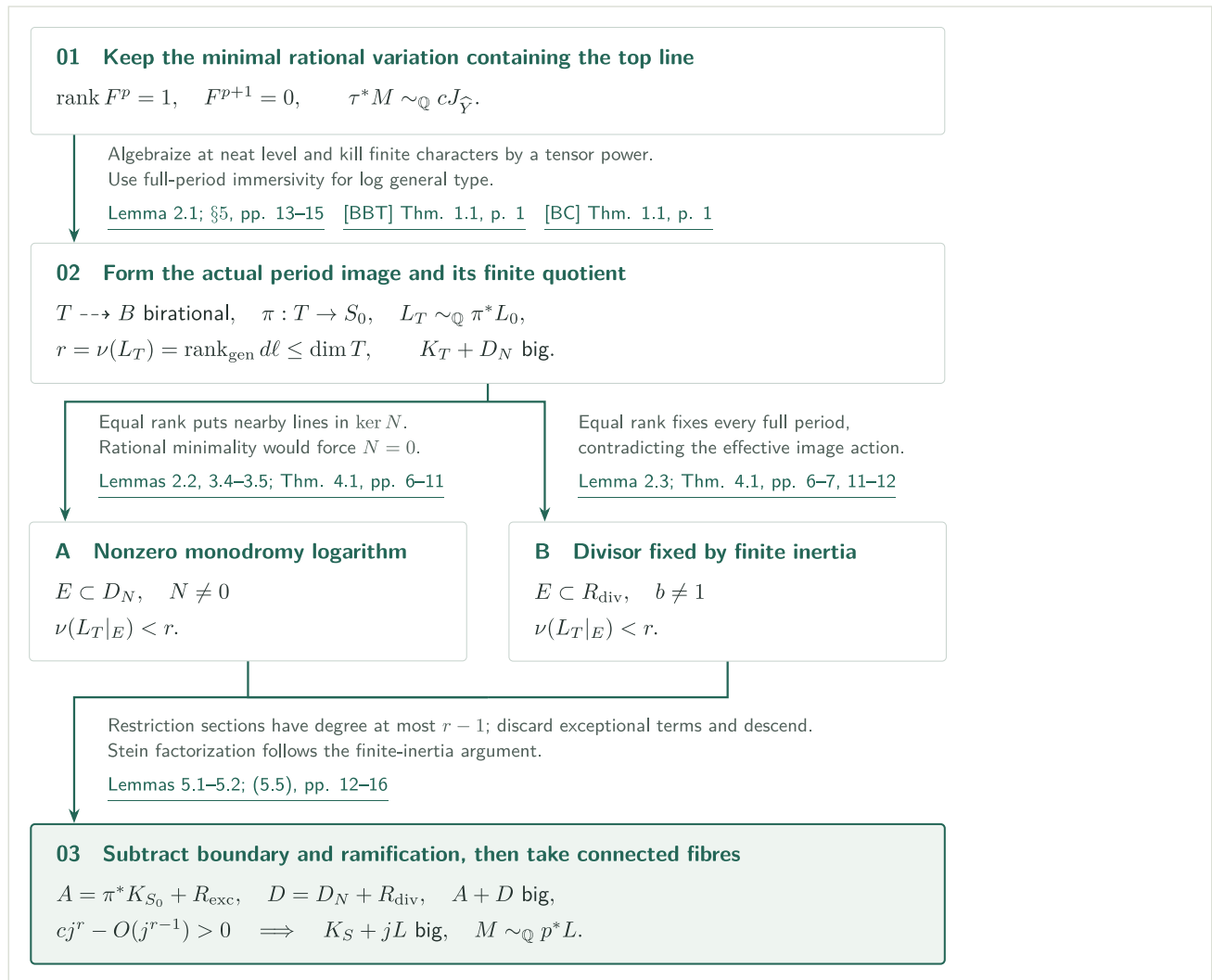
- [BBT](#) Bakker–Brunebarbe–Tsimmerman.
- [BC](#) Brunebarbe–Cadorel.
- [FM](#) Fujino–Mori.
- [Amb](#) Ambro.
- [Fuj](#) Fujino.
- [Has](#) Hashizume.

The [dependency table](#) records versions and checking scope. Citations link to the sources; page numbers for GitHub previews are given in the labels.

2. Theorem 1.2 — Removing boundary and ramification from the period image

Proof route. Two preparatory arguments meet in Theorem 1.1. This section establishes the adjoint positivity of Theorem 1.2; the canonical bundle formula supplies the section comparison. The final two-base argument and interpolation combine them to prove ordinary Iitaka subadditivity.

Let T be a smooth projective model of the actual full period image at neat level, J its extended highest line, and $L_T = cJ$. Write D_N for the boundary with nonzero monodromy logarithm, R_{div} for divisorial ramification over the finite quotient, and R_{exc} for the Jacobian components exceptional over the quotient base. The number $r = \nu(J)$ measures the highest-line map and need not equal $\dim T$.



Rational minimality and two rank drops yield adjoint bigness

Open figure ↗ TeX source ↓

1. Retain the finite action on the actual period image

Replace the variation by its smallest rational subvariation containing the highest line. A generic Hodge structure is then rationally generated by that line, allowing one to eliminate rational normal monodromy subgroups fixing it. Algebraize the period image at neat level and retain the function field of its normalization when taking the finite quotient. Tensor powers kill the finite stabilizer characters on the line, permitting rational line-bundle descent. The identity with the original M is obtained by comparing extensions on a common alteration and applying a norm; it is not inferred solely from the open locus.

Lemmas 2.1–2.3 · pp. 5–7; §5, (5.1) · pp. 13–15 · BBT, Theorem 1.1 · p. 1

2. Treat the nilpotent and finite obstructions separately

The nef line satisfies $\nu(J) = \text{rank}_{\text{gen}} d\ell$. If its numerical dimension did not drop on a divisor E , the tangential limiting map would have rank r . Lemma 3.4 would then put the nearby line image inside that limiting image. For $N \neq 0$, it lies in $\mathbb{P}(\ker N)$, contradicting rational minimality. For finite inertia it lies in a single eigenspace; Lemma 2.3 then forces every full period to be fixed. The final contradiction uses the fact that $\mathbb{C}(T)$ is the **function field of the actual period image**. Passing first to an arbitrary finite cover or Stein factor would lose this argument.

Proposition 3.2 · pp. 7–8; Lemmas 3.4–3.5 · pp. 9–10; Theorem 4.1 · pp. 11–12 ·
BBT, Lemma 6.17 · pp. 43–44

3. Remove both divisors by comparing degrees of section growth

After extending the variation across components with $N = 0$, generic immersivity of the full period map makes $K_T + D_N$ big. Apply Lemma 5.1 with

$$A = K_T - R_{\text{div}} = \pi^* K_{S_0} + R_{\text{exc}}, \quad D = D_N + R_{\text{div}}, \quad J = L_T.$$

An ample perturbation yields interior section growth of degree r in j , whereas restrictions to the removed divisors have degree at most $r - 1$. Thus $A + jL_T$ is big. Removing the exceptional part and descending through the generically finite map makes $K_{S_0} + jL_0$ big. Connected-fibre factorization is performed only afterwards; its effective Jacobian preserves bigness of $K_S + jL$.

Lemmas 5.1–5.2 · pp. 12–13; (5.2)–(5.5) · pp. 15–16 · BC, Theorem 1.1 · p. 1

The identity $M = p^*L$ and adjoint bigness on the base enter the weak-positivity and interpolation argument of Section 4. Semiamplessness of M is not concluded here.

PH, §7.3 · pp. 25–26

3. The canonical bundle formula — Preparing the line and the section comparison

Put $d = \kappa(F) \geq 0$ and write the relative Iitaka factorization as $\tilde{X} \xrightarrow{g} Y \xrightarrow{q} Z$. Then $\dim(Y/Z) = d$, and the geometric generic fibre C of g has Kodaira dimension 0. The construction must retain both the line required by Theorem 1.2 and its comparison with the original pluricanonical sections.

01 Take the relative Iitaka map and its Kodaira-zero fibre

$$\tilde{X} \xrightarrow{g} Y \xrightarrow{q} Z, \quad d = \dim(Y/Z) = \kappa(F), \quad \kappa(C) = 0.$$

Minimality gives a degree- b cover with a unique top form.
Keep the rational variation rather than only its character eigenspace.

[Lemma 6.3; Prop. 6.4, pp. 19–21](#)

02 Spread out the least-index root cover

$$b = \min\{m > 0 : H^0(C, mK_C) \neq 0\}, \quad \xi^b = s/\omega^b, \\ \kappa(W) = 0, \quad h^0(W, K_W) = 1, \quad \mathbb{V} = R^n h_* \mathbb{Q}, \quad \text{rank } F^n = 1.$$

Compare the bundle formula and Hodge eigensheaf on extended alterations.
The auxiliary boundary $-\text{div}(s)/b$ can be negative.

[Thm. 6.1; \(6.4\), pp. 16–20](#)

[\[FM\] Thm. 4.5, PDF pp. 11–12](#) [\[Amb\] Lemma 5.2; Prop. 5.5, pp. 15, 17–18](#)

03 Retain the extended Hodge identity and the residual divisor

$$D = K_Y + B + M, \quad B \geq 0, \quad (Y, B) \text{ klt}, \quad M \text{ nef}, \quad \tau^* M \sim_{\mathbb{Q}} aJ, \\ K_{\tilde{X}} \sim_{\mathbb{Q}} g^* D + R, \quad g_* \mathcal{O}(\lfloor kR^+ \rfloor) = \mathcal{O}_Y, \quad R^- \text{ exceptional over } X.$$

Multiply by the section of R^+ and use exceptionality of R^- .
On the generic fibre use all-degree direct images and flat base change.

[Lemma 6.5; Lemma 6.6, pp. 21–22](#)

04 Compare sections on the total space and the generic fibre

$$H^0(Y, mD) \hookrightarrow H^0(X, mK_X), \quad \kappa(X) \geq \kappa(Y, D), \\ d \leq \kappa(Y_{\bar{\eta}}, D|_{Y_{\bar{\eta}}}) \leq \dim Y_{\bar{\eta}} = d \implies D|_{Y_{\bar{\eta}}} \text{ big}.$$

The least-index root cover and residual divisor supply the line and section comparison

[Open figure ↗](#) [TeX source ↓](#)

1. Use the least index to obtain the entire rank-one top piece

Let $b = \min\{m > 0 : H^0(C, mK_C) \neq 0\}$. Its unique section s defines the root cover by $\xi^b = s/\omega^b$. Lemma 6.3 proves $\kappa(W) = 0$ and $h^0(W, K_W) = 1$ for a resolution W . Spreading out the construction makes the entire top piece of $R^n h_* \mathbb{Q}$ rank one. Even when the cyclic character is not rational, the full rational variation is retained.

Lemma 6.3, (6.4) · pp. 19–20

2. Separate the negative auxiliary boundary from the residual divisor

With $B \geq 0$, (Y, B) klt, and M nef, the formula gives

$$K_{\tilde{X}} \sim_{\mathbb{Q}} g^*(K_Y + B + M) + R, \quad g_*\mathcal{O}_{\tilde{X}}(\lfloor kR^+ \rfloor) = \mathcal{O}_Y \quad (k > 0).$$

The image of R^- has codimension at least two on the base, and R^- is also exceptional over the original X . The latter property requires flattening; it does not follow just from the image dimension. On the generic fibre the auxiliary sub-pair has boundary $-\operatorname{div}(s)/b$, which need not be effective. The argument uses nefness and the extended Hodge-line identification, without importing moduli abundance that requires an effective generic boundary.

Theorem 6.1, Remark 6.2 · pp. 16–19 · FM, Theorem 4.5 · PDF pp. 11–12 ·
Amb, Lemma 5.2(4)–(5), Proposition 5.5 · pp. 15, 17–18

3. Compare sections on the total space and on the generic fibre

Set $D = K_Y + B + M$. Exceptionality of R^- identifies the sections of $K_{\tilde{X}} + R^-$ with the original pluricanonical sections. Multiplication by the section of R^+ injects $H^0(Y, mD)$ into that space. On the geometric generic q -fibre, flat base change and the all-degree direct-image condition give

$$\kappa(X) \geq \kappa(Y, D), \quad d = \kappa(F) \leq \kappa(Y_{\bar{\eta}}, D|_{Y_{\bar{\eta}}}) \leq \dim Y_{\bar{\eta}} = d.$$

Thus $D|_{Y_{\bar{\eta}}}$ is big. Subsequent birational modifications use the positive part of the crepant boundary and an effective exceptional correction to preserve these comparisons.

Proposition 6.4, Lemmas 6.5–6.6 · pp. 20–22

Section 4 uses both q to the original base and p to the period base. The section injection is retained to transfer the final lower bound back to X .

PH, Proposition 6.4 · §7.3 · pp. 20–21, 25–26

4. Theorem 1.1 — Two bases and an interpolation identity

Assume $\kappa(Z) \geq 0$. The identity $M = p^*L$ from Theorem 1.2 does not supply a factorization between $p : Y \rightarrow S$ and $q : Y \rightarrow Z$. If S is a point, apply log-general-type-fibre subadditivity directly to q . The remaining argument treats $\dim S > 0$.

01 Keep the two bases distinct

$$q : Y \rightarrow Z, \quad p : Y \rightarrow S, \quad D = K_Y + B + M, \quad M \sim_{\mathbb{Q}} p^* L, \\ \kappa(Z) \geq 0, \quad D|_{Y_{\bar{\eta}}} \text{ big}, \quad K_S + jL \text{ big } (j \gg 0).$$

Contract forms from Z to W_s . The general fibre of $Y_s \rightarrow W_s$ is of log general type, so apply the established addition theorem.

[Lemma 7.3, pp. 23–24](#) [\[Has\] Thm. 2.11, p. 5](#)

02 Use the product map for nonvanishing on a period fibre

$$Y \rightarrow W \rightarrow S \times Z, \quad \kappa(W_s) \geq 0, \\ \kappa(Y_s, K_{Y_s} + B|_{Y_s}) \geq 0.$$

Apply weak positivity to a positive-rank log pluricanonical direct image. Multiply symmetric powers and push down exceptional poles.

[Lemma 7.4; \(7.6\), pp. 24–26](#) [\[Fuj\] Thm. 1.1, pp. 1–2](#)

03 Construct an effective divisor using weak positivity

$$P_0 = K_S + jL - H \text{ big}, \quad E_1 = K_Y + B + jM - p^* H \sim_{\mathbb{Q}} E \geq 0.$$

Choose v by openness of bigness and a general Γ using ample A_S . Then E_2 is q -generically big and $(Y, B + \Gamma)$ is klt.

[\(7.5\)–\(7.7\), p. 26](#)

04 Interpolate coefficients and return to an effective klt pair

$$0 < v < 1, \quad \lambda = \frac{1-v}{j-v}, \quad A_S = vL + \frac{\lambda}{1-\lambda} H, \\ E_2 = K_Y + B + p^* A_S \sim_{\mathbb{Q}} K_Y + B + \Gamma, \quad D \sim_{\mathbb{Q}} \lambda E_1 + (1-\lambda) E_2.$$

Apply addition to q and multiply by sections of E_1 . Scalar extension preserves all three Kodaira dimensions.

[\[Has\] Thm. 2.11, p. 5](#) [§7.3; Lemma 7.5, pp. 26–27](#)

END Apply addition, transfer sections, and extend scalars

$$\kappa(X) \geq \kappa(Y, D) \geq \kappa(Y, E_2) \geq \kappa(F) + \kappa(Z), \\ H^0(V, nK_V) \otimes_K K' \simeq H^0(V_{K'}, nK_{V_{K'}}).$$

The product map, weak positivity, exact interpolation, and scalar extension

[Open figure ↗](#) [TeX source ↓](#)

1. Obtain nonvanishing on a period fibre through the product map

Resolve the Stein factorization of the image of (p, q) as $Y \rightarrow W \rightarrow S \times Z$. A general W_s is generically finite onto its image in Z , while $W \rightarrow Z$ is dominant. Contracting top forms with the exterior base-direction factor turns a pluricanonical section on Z into a nonzero section on W_s . On the general fibre of $Y_s \rightarrow W_s$, the line M becomes trivial, so bigness of D becomes log-canonical bigness. A first use of log-general-type-fibre subadditivity gives log nonvanishing on the geometric generic p -fibre.

[Lemma 7.3 · pp. 23–24](#) · [Has, Theorem 2.11 · p. 5](#)

2. Use weak positivity to construct an effective divisor with a negative ample twist

Choose $j > 1$ and an ample rational divisor H with $K_S + jL - H$ big. The preceding nonvanishing makes a relative log pluricanonical direct image have positive rank, allowing weak positivity. Multiplying sections of symmetric powers and pushing down exceptional poles after flattening yields

$$E_1 := K_Y + B + jM - p^*H \sim_{\mathbb{Q}} \text{an effective divisor.}$$

The extension and pushdown in Lemma 7.4 connect generic generation to effectivity on the whole total space.

Theorem 7.2, Lemma 7.4, (7.6) · pp. 23–26 · Fuj, Theorem 1.1 · pp. 1–2

3. Balance coefficients and reduce to an effective klt pair

Openness of the big cone allows $0 < v < 1$ with $(K_Y + B + vM)|_{Y_{\bar{\eta}}}$ big. Set

$$\lambda = \frac{1-v}{j-v}, \quad A_S = vL + \frac{\lambda}{1-\lambda}H, \quad E_2 = K_Y + B + p^*A_S, \quad D \sim_{\mathbb{Q}} \lambda E_1 + (1-\lambda)E_2.$$

These choices simultaneously balance the coefficient of M and cancel the ample twist. Since A_S is ample, a sufficiently divisible general section gives $\Gamma \sim_{\mathbb{Q}} p^*A_S$ with $(Y, B + \Gamma)$ klt. The divisor E_2 is its log canonical class and is big on the generic q -fibre. Apply subadditivity a second time and multiply by sections of E_1 to obtain

$$\kappa(X) \geq \kappa(Y, D) \geq \kappa(Y, E_2) \geq d + \kappa(Z).$$

Finally descend the data to a finitely generated field $K/\mathbb{Q}$ and choose $K \hookrightarrow \mathbb{C}$. Flat base change for every pluricanonical space, including those of the geometric generic fibre, transfers the result to any algebraically closed characteristic-zero field.

§7.3, (7.5)–(7.7) · pp. 25–26; Lemma 7.5, §7.4 · p. 27

This yields ordinary subadditivity, Theorem 1.1. WV’s alternative adjoint comparison concerns the separate results of §8; this main theorem is not substituted as a required WV input.

PH, Theorem 1.1 · p. 2; Theorem 8.1 · p. 28 · WV, Remark 3.17 · p. 25

5. Which papers supply which steps

Input

BBT arXiv:1811.12230v3, Theorem 1.1 / Lemma 6.17 (pp. 1, 43–44)

Use and role

Period image in §5; nefness and numerical rank in §3

Checking scope

Statements compared with uses

Input

BC arXiv:1707.01327v1, Theorem 1.1 (p. 1)

Use and role

Bigness of $K_T + D_N$ in §5 p.15

Checking scope

Statement, immersivity, and choice of boundary compared

Input

FM JDG 56 (2000), Theorem 4.5 (PDF pp. 11–12; journal pp. 177–178)

Use and role

Residual divisor and sections in Theorem 6.1

Checking scope

Statement and boundary-zero use compared; complete model construction unchecked

Input

Amb arXiv:math/0210271v1, Theorem 4.4, Lemma 5.2(4)–(5), Proposition 5.5 (pp. 12–15, 17–18)

Use and role

Hodge eigensheaf and pullback in Theorem 6.1 and (6.4)

Checking scope

Relevant statements compared; full cited package and base-change proofs unchecked

Input

Fuj 2015-06-30, v0.54, Theorem 1.1 (pp. 1–2)

Use and role

Theorem 7.2 → effectivity in Lemma 7.4

Checking scope

Statement and application to the projective klt pair compared

Input

Has arXiv:1902.10923v2, Theorem 2.11 (p. 5)

Use and role

Addition in Lemma 7.3 and §7.3

Checking scope

Statement compared; no abundance of the base canonical class required

These are direct inputs to the main route. The uses of the fixed-part theorem, monodromy normality, and nilpotent-orbit and extension theory were read in PH, but their external source statements have not all been checked. Within catalogue 033, WV, Remark 3.17 · p. 25 cites Lemma 8.4 / Theorem 8.1 as an alternative adjoint comparison. WV explicitly uses its direct comparison in its own proof, so this relationship is `alternative`. The ordinary-subadditivity comparison with OI is likewise treated as a separate proof route; no direct PH→OI edge is asserted. Other relationships remain unexamined.

6. Return to the sources

- §§2–5 · pp. 5–16: rational minimality, tangential limits, the two rank losses, quotient descent, and volume estimates.
- §6 · pp. 16–22: the least-index root cover, the Hodge identification across the boundary, and residual section comparisons.
- §7 · pp. 22–27: the product map, effectivity from weak positivity, interpolation, and scalar extension.
- §§8–9 · pp. 27–39: alternative boundary estimates by integral tensor replacement and Thom forms, separate from the main route.

The reading covers Theorems 1.1–1.2, Lemmas 2.1–2.3 / 3.4, Theorem 4.1, descent and volume estimates in §5, and the central proof passages of §§6–7. Section 5 specifies which input statements and applications were compared. Only the statements of Theorem 8.1 / Lemma 8.4 and the corresponding WV comparison were read. The boundary mixed-Hodge identification, multivariable extensions, every auxiliary proof, the alternative proofs in §§8–9, and full external proofs have not been independently verified. This is an AI-generated account of the authors' argument, not certification of the manuscript or expert peer review. The source is the 40-page version of 2026-09-27, pinned to commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

B-semiampleness for compact log-smooth Kähler fibrations

From threshold orders to global generation through a family of product decompositions

The manuscript claims stabilization and semiampness of the moduli b-line for compact log-smooth Kähler fibrations, allowing boundary coefficients equal to one. Its central steps identify the extension order of a local root eigenline with the log canonical threshold, then descend global generation of a specified holomorphic line bundle from a product family over a compact auxiliary base.

Manuscript author: OpenAI · 2026-09-10 · 38 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

The following is the manuscript's claim. Spaces and maps are complex analytic. Equality in $\text{Pic}(X)_{\mathbb{Q}} = \text{Pic}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$ means an isomorphism of holomorphic line bundles after a common positive integral multiple.

Let $f : Y \rightarrow X$ be a surjective holomorphic map with connected fibres between smooth compact connected Kähler manifolds. Let Δ be an effective rational SNC divisor with coefficients in $[0, 1]$, and assume

$$K_Y + \Delta \sim_{\mathbb{Q}} f^*L, \quad L \in \text{Pic}(X)_{\mathbb{Q}}.$$

For a smooth compact Kähler modification $\mu : X' \rightarrow X$, resolve the main component to obtain $f' : Y' \rightarrow X'$ and $h : Y' \rightarrow Y$, and set $\Delta' = h^*\Delta - K_{Y'/Y}$. For a prime divisor P on X' , define

$$t_P = \inf_{E \rightarrow P} \frac{a(E; Y', \Delta')}{\text{ord}_E((f')^*P)}, \quad B_{X'} = \sum_P (1 - t_P)P, \quad M_{X'} = \mu^*L - K_{X'} - B_{X'}.$$

The infimum ranges over all divisorial valuations dominating P with positive denominator; a is log discrepancy. Exceptional coefficients of Δ' may be negative. This is the threshold discriminant, distinct from the inf-multiplicity divisor of an orbifold base.

§1.1, (1.1)–(1.3) · pp. 2–3

Theorem 1.1 — Stabilization and global generation

Under these assumptions there is a smooth compact Kähler modification $\mu_0 : S \rightarrow X$ with the following properties.

(a) For every smooth compact Kähler modification $\nu : S_1 \rightarrow S$,

$$M_{S_1} = \nu^* M_S \quad \text{in } \text{Pic}(S_1)_{\mathbb{Q}}.$$

(b) For some positive integer m , the actual holomorphic line bundle representing mM_S is globally generated.

The statement includes a point base and relative dimension zero. No projectivity of f, X, Y is assumed.

Theorem 1.1 · p. 3

References on the arrows

Unprefixed result numbers refer to [BS](#). The main external inputs use the following keys.

- [FF](#) Fujino–Fujisawa.
- [MWWZ](#) Matsumura–Wang–Wu–Zhang.
- [BFMT](#) Bakker–Filipazzi–Mauri–Tsimmerman.
- [Toma](#) Toma.

The [dependency table](#) records the checks. Diagram citations link to the sources; GitHub PDF page numbers are specified in the labels.

2. Overview of Theorem 1.1

Proof route. First, compare thresholds with extension orders to obtain stabilization in Theorem 1.1(a). Part (b) then organizes product decompositions into a family, obtains semiampleness for each factor, and extends the specified isomorphism before descending global generation.

Retain the actual moduli line stabilized in (a) and obtain global generation at every point in (b). The two main parts are the threshold calculation and the product/norm comparison on an auxiliary base P .



Overview from threshold stabilization through product factors to extension and descent

Open figure ↗ TeX source ↓

1. Fix the stabilized line by a local calculation

Compare the root-eigenline extension order with the lct and obtain pullback identities on every higher model. Track the specified holomorphic line, not only its numerical class.

Theorem 4.4, Corollary 4.5 · pp.15–17

2. Realize a product family and handle the factors separately

On a compact auxiliary base, separate the projective factor from the torus and symplectic factors. The former uses b-semiampleness for a projective comparison family; the latter use arithmetic period compactifications. Retain the same specified line and product norm.

Proposition 5.6, Proposition 6.1, Proposition 7.5 · pp.22–34

3. Extend the specified isomorphism and descend generation everywhere

Two-sided norm bounds remove poles from both the isomorphism and its inverse, leaving neither boundary nor flat twists. Norms after the Stein factorization of a proper surjection give generation by one common multiple.

Lemmas 8.1, 8.3, §9 · pp.34–37

Together (a) and (b) prove Theorem 1.1. The detailed diagrams below expand stabilization, family construction, semiampleness of the factors, and descent of the isomorphism and sections.

3. Theorem 1.1(a) — Read thresholds as extension orders

Let $p : W \rightarrow S$ be a model with SNC boundary, let D_W be the crepant boundary, and put $Q_S = \mu_0^* L - K_S$. Fix $\mathcal{O}_W(m(K_{W/S} + D_W)) \simeq p^* \mathcal{O}_S(mQ_S)$ for sufficiently divisible m . The cover obtained by taking an m th root of a local generator retains every component, even if disconnected.

1 Take a local root and retain every sheet

$$K_{W/S} + D_W \sim_{\mathbb{Q}} p^* Q_S, \quad F^d \mathbb{V}_{\chi} = \mathcal{O}_{\tau}$$

The specified character eigenline; not the entire highest space

Use the maximal coefficient-one horizontal intersection and iterated residue.
Prop. 3.4, Lem. 3.5, pp. 10–12

2 Pass to a pure line by the deepest residue

$$F^d \operatorname{Gr}_{d+k}^W \mathbb{V}_{\chi} \simeq \text{klt log-volume line on a depth-}k \text{ stratum}$$

Retain the local frame and its degree- m identification

Test extension in both directions on disk models, including the Jacobian of du .
Lem. 4.2–4.3, Thm. 4.4, pp. 13–16

Mixed Hodge extension and highest direct image: [FF] Thm. 1.1, p. 2

3 Identify threshold and extension order

$$t_P = \min_E \frac{1 - \delta_E}{a_E}, \quad \operatorname{ord}_{\operatorname{Hdg}} \tau = t_P - 1$$

$$a_E = \operatorname{ord}_E(t), \quad \delta_E = \operatorname{coeff}_E D_W$$

Kill boundary orders on power charts; cancel relative canonical and threshold changes.
Thm. 4.4, Cor. 4.5, pp. 15–17

4 Correct the boundary and descend to every higher model

$$-K_{S_1/S} - B_{S_1} + \nu^* B_S = 0$$

$$M_{S_1} = \nu^* M_S \quad \text{in } \operatorname{Pic}(S_1)_{\mathbb{Q}}$$

Theorem 1.1(a), including centres inside the good open set

The root eigenline, residues, threshold orders, and stabilization on higher models

[Open figure ↗](#) [TeX source ↓](#)

1. Isolate the relevant line in the mixed Hodge structure

For the character χ of the tautological root τ , the line $F^d \mathbb{V}_{\chi} = \mathcal{O}_{\tau}$ has rank one. This does not assert rank one for the highest space of the entire variation. Let k be the largest number of horizontal coefficient-one components meeting on a good fibre. The line lies in $\operatorname{Gr}_{d+k}^W$, and after removing the Tate twist, iterated residue identifies it with the pure eigenline of a klt log-volume on a depth- k stratum. The local frame and its degree- m identification are retained. If $k = 0$, no residue operation is needed.

Proposition 3.4, Lemma 3.5 · pp. 10–12

2. Determine the extension order in both directions

Above $P = (t = 0)$, put $a_E = \text{ord}_E(t)$ and $\delta_E = \text{coeff}_E D_W$. The minimum on an SNC resolution equals the threshold defined using all divisorial valuations. The manuscript uses normalized root models over a disk and mixed Hodge extensions to prove

$$t_P = \min_E \frac{1 - \delta_E}{a_E}, \quad \text{ord}_{\text{Hdg}} \tau = \min_E \frac{l_E + 1 - a_E}{a_E} = t_P - 1, \quad l_E = -\delta_E.$$

The Hodge order is normalized by dividing by the ramification index of the base change. After $t = u^N$, test $u^{-N\delta} \tau$. Transforming the total form introduces the Jacobian of du . Extension holds when every $\alpha_E = l_E + 1 - a_E(1 + b)$ is nonnegative; a negative value produces a pole on a retained component over E . Thus the calculation establishes necessity as well as sufficiency. The direct-image identification from FF applies to disk models whose strata dominate the base and are Kähler. Dualizing the lower compact-support extension gives the upper highest-line extension in ordinary cohomology.

Lemmas 4.1–4.3, Theorem 4.4 · pp. 12–16 · FF, Theorem 1.1 · p. 2

3. Glue the boundary correction as an actual line bundle

If $B_S = \sum b_i(t_i = 0)$, the corrected frame $\prod_i w_i^{N b_i} \pi^* \tau$ has order zero on a power chart $t_i = w_i^{N_i}$. A tensor power removes roots-of-units ambiguities, and Hartogs extension handles crossings. The prescribed identification extends to $\pi^* \mathcal{O}_S(amM_S) \simeq E_X^{\otimes am}$. On a higher model, retain $Q_{S_1} = \nu^* Q_S - K_{S_1/S}$ in the same calculation. Relative canonical orders cancel the change of thresholds, yielding

$$-K_{S_1/S} - B_{S_1} + \nu^* B_S = 0.$$

This includes modifications with centres inside the good open set and proves (a).

Theorem 4.4, Corollary 4.5 · pp. 15–17

Retain the stabilized moduli line and its specified log-volume identification; this fixes the object compared by the ensuing product decomposition.

BS, Corollaries 4.5–4.6 · pp. 16–18

4. Preparing Theorem 1.1(b) — Turn fibrewise decompositions into a family

The deepest stratum supplies a family with klt log Calabi–Yau good fibres. Write G_m for its degree- m log-plurivolume line. Beyond a decomposition of each fibre, the proof needs a single family carrying both a line identification and its norm.

1 Extract a klt log-volume family with a section

$$\Sigma \longrightarrow S, \quad m(K_F + D_F) \sim 0, \quad (F, D_F) \text{ klt}$$

Stein factorization of the deepest stratum and a diagonal section

Combine fibrewise splitting with finiteness of subgroups of fixed index.

Lem. 5.1, 5.3–5.4, pp. 18–21

Splitting input: [MWWZ] Cor. 1.3, pp. 2–3

2 Enumerate finite étale product decompositions in families

$$(J, D_J) = (Q, C) \times \prod_i T_i$$

Q is projective; each T_i is a torus or irreducible symplectic factor.

Impose constructible graph/boundary conditions; apply Baire to countably many closures.

Lem. 5.5, Prop. 5.6, pp. 22–25

Compact Douady components: [Toma] Cor. 5.3, p. 19

3 Select a dominating compact parameter space

$$K = B \times \text{Hilb} \times \prod_i \text{Douady}_i \times \text{Douady}_{\text{graph}}$$

$$r : P \longrightarrow S \quad \text{proper, surjective}$$

The auxiliary base P is smooth, compact, and complex.

Carry the log-volume isomorphism and residue-integral comparison with the family.

Prop. 5.6, (5.5)–(5.6), pp. 22–25

4 Retain the specified line and product norm

$$r^* G_m \simeq G_{m,Q} \otimes \bigotimes_i \lambda_i^{\otimes m} \quad \text{on } P^\circ$$

$$\|s\|_{\text{res}} = c \|s_Q\|_{Q,\text{res}} \prod_i \|\omega_i\|_{\text{Hdg}}^m, \quad c > 0$$

No Kählerness, projectivity, or generic finiteness is assumed for P .

Finite covers and compact parameter spaces produce a product family

[Open figure ↗](#) [TeX source ↓](#)

1. Enumerate product decompositions as families of finite covers

Stein factorization of the deepest stratum and resolution of a graph produce a family with a diagonal section. The finite étale decomposition supplied by [MWWZ](#) separates each klt fibre into a projective factor (Q, C) and torus or irreducible holomorphic symplectic factors T_i . All boundary lies on Q , which may be a point. The section expresses the fundamental group as a semidirect product. Finiteness of subgroups of fixed index in the finitely generated fibre group then spreads each finite étale cover after a finite base cover.

[Lemmas 5.1, 5.3–5.4 · pp. 18–21](#) · [MWWZ, Corollary 1.3 · pp. 2–3](#)

2. Use compact parameter spaces and Baire to choose one family

Hilbert schemes record the projective factor; Douady components in fixed compact Kähler spaces record the T_i and the product graph. Graph, étaleness, and boundary conditions are imposed as analytically constructible conditions. Torus/symplectic type, expressed by Hodge numbers and cup-product ranks, persists along a stratum containing one point of the desired type. Countably many compact closures have images covering the good base, so Baire selects a dominating one. Its resolution gives a proper surjection $r : P \rightarrow S$. The smooth compact complex manifold P is not assumed Kähler, projective, or generically finite over S .

Lemma 5.5, Proposition 5.6 · pp. 22–25 · Toma, Corollary 5.3 · PDF p. 19

3. Carry the specified line and norm along the decomposition

Over a dense open set P° , the chosen finite étale cover is a family satisfying

$$(J, D_J) = (Q, C) \times_{P^\circ} \prod_i T_i, \quad r^* G_m \simeq G_{m,Q} \otimes \bigotimes_i \lambda_i^{\otimes m},$$

where λ_i is the highest line of the corresponding factor. The residue-integral norm agrees with the product norm up to a positive constant depending only on the cover degree. This **specified isomorphism**, together with its norm, is passed to the next steps; a comparison only of numerical classes or up to an unspecified flat twist would not suffice.

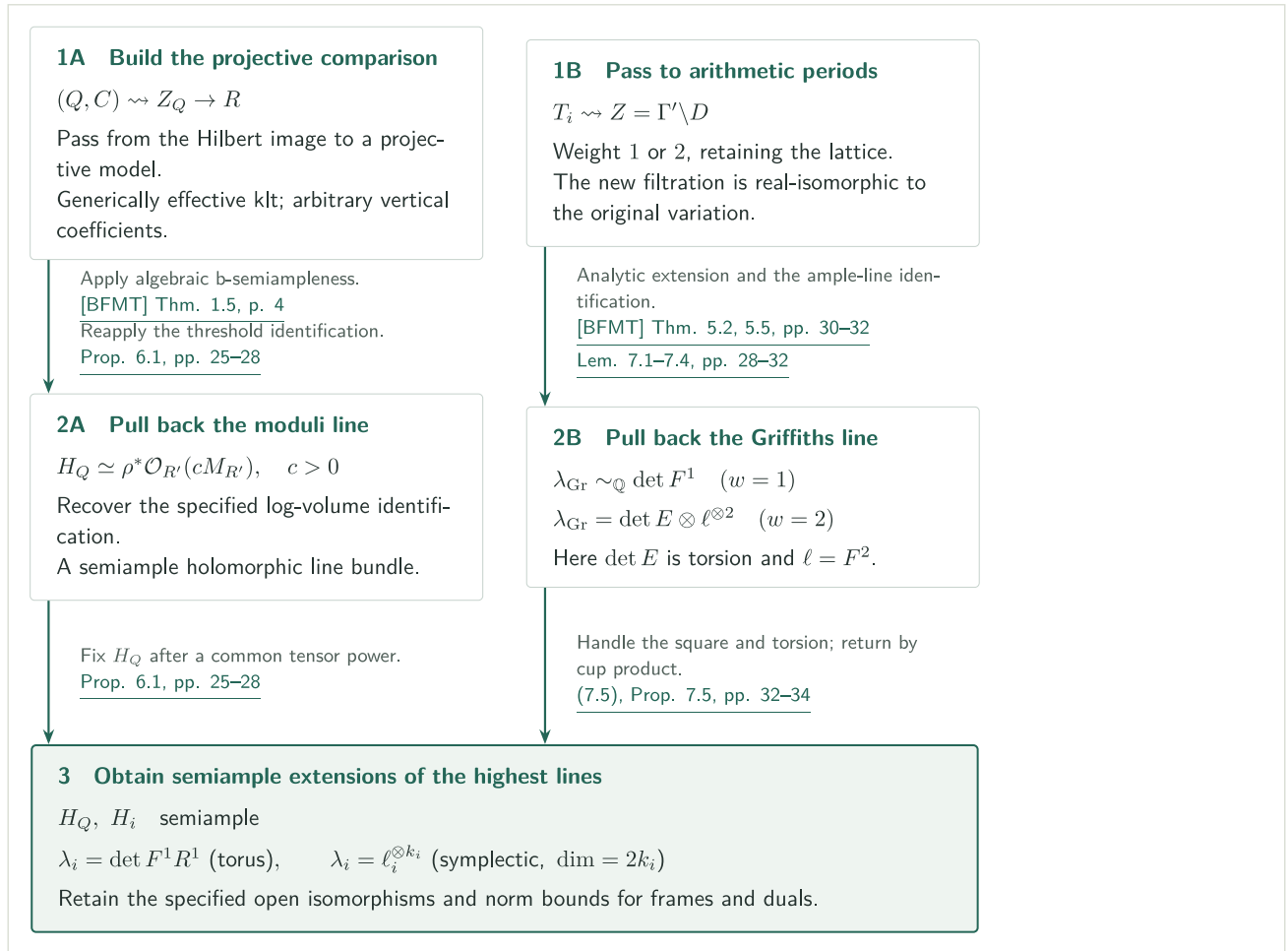
Corollary 4.6 · pp. 17–18; Proposition 5.6, (5.5)–(5.6) · pp. 22–25

The product decomposition and product norm on the same auxiliary base P enter the separate treatments of the projective and torus/symplectic factors.

BS, Proposition 5.6 · §§6–7 · pp. 22–34

5. Obtain semiample extensions from the factors

Different inputs treat the projective pair (Q, C) and the possibly nonprojective T_i . In this section, P also denotes the auxiliary base after finite covers, graph resolutions, and a common tensor power.



Separate projective comparison from the period maps of torus and symplectic factors

[Open figure ↗](#) [TeX source ↓](#)

1. Apply algebraic b-semiampleness to the projective factor

The compact image of $\mathbf{P}$ in a Hilbert scheme is projective by Chow. The universal family over a projective model $\mathbf{R}$ yields a projective lc-trivial fibration with generically effective klt boundary. Vertical coefficients in the auxiliary boundary defined by a rational frame may be negative. Pull back the line supplied by algebraic b-semiampleness in [BFMT](#), then reapply the manuscript's local threshold calculation to recover the specified log-volume identification and two-sided norm estimates. This does not require a global rational frame on the original nonprojective space.

[Proposition 6.1 · pp. 25–28](#) · [BFMT, Theorem 1.5, Definition 6.18, Theorem 6.28 · pp. 4, 42, 44](#)

2. Replace the real polarization and pass to arithmetic period quotients

Since the T_i occur inside a fixed compact Kähler space, its Kähler class supplies flat real polarizations. Lemma 7.1 uses a nearby rational monodromy-invariant form and a commuting real automorphism to change the filtration while retaining the lattice. It does not make the original fibres rationally polarized, but a real isomorphism compares the holomorphic Hodge

bundles and their extensions. Weight 1 for tori and weight 2 with $h^{2,0} = 1$ for symplectic factors give maps to neat Siegel or type IV arithmetic quotients. Analytic extension of the period map, with the Griffiths-line identification, pulls back sections of an ample line on the compactification.

Lemmas 7.1–7.4 · pp. 28–32 · BFMT, Theorems 5.2, 5.5 · pp. 30–32

3. Recover the required highest lines from Griffiths lines

For a torus, $\lambda_i \simeq \det F^1 R^1$. For a symplectic factor of dimension $2k_i$, put $\ell_i = F^2 R^2$; cup product gives $\lambda_i \simeq \ell_i^{\otimes k_i}$. The weight-2 Griffiths line is $\det E \otimes \ell_i^{\otimes 2}$. Its square and the finite-order factor $\det E$ must be handled before concluding semiample of ℓ_i . A common finite cover treats all monodromies; compatibility of extensions with tensor and cup products yields semiample extensions H_Q, H_i of the specified lines.

Lemma 7.3, §7.4, Proposition 7.5, (7.5) · pp. 30, 32–34

Tensor the semiample extensions and retain the two-sided norm bounds needed to extend the specified isomorphism across the boundary.

BS, Lemma 8.1 · §9 · pp. 34–37

6. Theorem 1.1(b) — Extend the specified isomorphism and descend sections

The preceding identifications initially live on P° . First extend them without a residual boundary twist; then descend global generation through a proper surjection.

1 Bound the comparison and its inverse

$$g : r^* \mathcal{O}_S(NM_S)|_{P^\circ} \simeq (H_Q \otimes \bigotimes_i H_i)|_{P^\circ}$$

$$|g|, |g^{-1}| \leq C(1 + |\log |t||)^b \quad \text{on transverse disks}$$

Two-sided bounds kill negative Laurent coefficients; Hartogs handles intersections.
Lem. 8.1, pp. 34–35; §9, pp. 36–37

2 Extend the specified isomorphism across the boundary

$$r^* \mathcal{O}_S(NM_S) \simeq H_Q \otimes \bigotimes_i H_i \quad \text{on } P$$

The right-hand side is semiample; no residual boundary or flat twist remains.

Descend sections along connected fibres; choose one section nonzero on a finite fibre.
Lem. 8.3, pp. 35–36

3 Use Stein factorization and the finite norm

$$P \xrightarrow{h} T \xrightarrow{q} S, \quad h_* \mathcal{O}_P = \mathcal{O}_T, \quad \deg q = d$$

$$q^* A^{\otimes n} \text{ globally generated} \implies A^{\otimes nd} \text{ globally generated}$$

Use one exponent at all points; generic finiteness of $P \rightarrow S$ is unnecessary.
Lem. 8.3; §9, pp. 35–37

4 Obtain global generation of the moduli line

$$A = \mathcal{O}_S(NM_S), \quad \boxed{\mathcal{O}_S(ndNM_S) \text{ globally generated}}$$

Theorem 1.1(b): a conclusion about the actual holomorphic line bundle

Two-sided norm bounds extend the isomorphism; Stein factorization and norms descend generation

[Open figure ↗](#) [TeX source ↓](#)

1. Exclude poles of both the isomorphism and its inverse

Corollary 4.6 and Propositions 6.1 and 7.5 bound norms of local frames and their duals by powers of logarithms. The product-norm identity therefore bounds the coefficient g of the comparison and g^{-1} on transverse disks by logarithmic powers. Negative Laurent coefficients vanish, so both extend holomorphically. Holomorphic dependence in tangential directions and Hartogs extension handle intersections. For a common sufficiently divisible $N > 0$, this gives an actual holomorphic line-bundle isomorphism

$$r^* \mathcal{O}_S(NM_S) \simeq H_Q \otimes \bigotimes_i H_i.$$

The right-hand factors absorb the required common tensor powers. Extending the specified open isomorphism also excludes an invisible flat twist.

[Lemma 8.1 · pp. 34–35; §9 · pp. 36–37](#)

2. Descend generation by norms without generic finiteness

Write the Stein factorization as $P \xrightarrow{h} T \xrightarrow{q} S$ with $\deg q = d$. Since $h_*\mathcal{O}_P = \mathcal{O}_T$, sections descend to T . If $q^*A^{\otimes n}$ is globally generated, for every $s \in S$ choose a single section nonzero at all points of the finite set $q^{-1}(s)$. Its norm is a section of $A^{\otimes nd}$ nonzero at s . Even if q is not flat, the generic product extends by normality. The same n, d work at every point, so A is semiample. Apply this to $A = \mathcal{O}_S(NM_S)$ to obtain (b).

Lemma 8.3 · pp. 35–36; §9 · pp. 36–37

One common multiple is generated at every point of S . Together with pullback stability in (a), this proves semiampleness of the moduli b-line. No direct dependency with another 033 paper is inferred from this conclusion.

BS, Theorem 1.1 · §9 · pp. 3, 36–37

7. Which papers supply which steps

Input

FF, Theorem 1.1, v3, p. 2

Use and role

§3 and Lemma 4.2: mixed Hodge extension and highest direct image for SNC families with Kähler strata

Check performed

Statement and application model compared

Input

MWWZ, Corollary 1.3, v1, pp. 2–3

Use and role

Lemma 5.3: finite étale splitting of effective klt fibres with numerically trivial log canonical class

Check performed

Statement and hypotheses compared

Input

Toma, Corollary 5.3, v3, PDF p. 19

Use and role

Proposition 5.6: compact Douady components in fixed compact Kähler spaces

Check performed

Statement and use compared

Input

BFMT, Theorem 1.5 / Definition 6.18 / Theorem 6.28, v2, pp. 4, 42, 44

Use and role

Proposition 6.1: projective comparison and Hodge/threshold moduli

Check performed

Generic effectivity, rank condition, and use compared

Input

BFMT, Theorems 5.2, 5.5, v2, pp. 30–32

Use and role

Lemma 7.4: analytic pullback of an ample Griffiths extension

Check performed

Statements and local liftability compared

These are direct inputs to auxiliary results in BS. Their complete proofs were not verified. Uses of Baily–Borel algebraicity, Fujiki countability, compactification of finite covers, and general pure Hodge extension theory were read in BS, but the corresponding external passages remain unchecked. The statement on p. 3 that the Campana orbifold Iitaka theorem is not used does not exclude other external dependencies. Direct dependencies involving other Catalog 033 papers or Catalog 034 remain uninvestigated.

8. Return to the sources

The source is OpenAI's September 10, 2026 manuscript, 38 pages, pinned at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Follow §§3–4, pp. 6–18 for eigenlines and thresholds, Proposition 5.6, pp. 22–25 for the family construction, and §§6–9, pp. 25–37 for factor semiampleness and descent.

The editorial reading covered the setup and both conclusions of Theorem 1.1, the cited proof passages, and the listed external statements at their applications. It does not certify the complete proof. The construction of mixed Hodge extensions, all details of the Hilbert–Douady parameter spaces, and the change of real polarization with all extension compatibilities were not independently proved. Connections whose external sources remain unchecked are left pending and distinguished from the author's claims.